

BANK OF THAILAND

Monetary Policy Report

March 2016

Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
March 2016

Mr. Veerathai Santiprabhob	Chairman
Mr. Mathee Supapongse	Vice Chairman
Mr. Paiboon Kittisrikangwan	Member
Mr. Jamlong Atikul	Member
Mr. Porametee Vimolsiri	Member
Mr. Sethaput Suthiwart-Narueput	Member
Mr. Apichai Boontherawara	Member

Monetary Policy in Thailand

Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprises the governor and two deputy governors, as well as four distinguished external members representing various sectors of the economy, with the aim of ensuring that monetary policy decisions are effective and transparent.

Monetary Policy Objective

The MPC sets monetary policy to promote the objective of supporting sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

Monetary Policy Target

The Cabinet approved the annual average headline inflation target of 2.5 $\pm$ 1.5 percent as the target for 2016 and for the medium term. The inflation target is to assure the general public that the MPC will take necessary policy actions to return headline inflation to the target within an appropriate time horizon without jeopardizing growth and macro-financial stability. In the event that headline inflation deviates from the target, the MPC shall explain the reasons behind the target breach to the Minister of Finance and the public, together with measures taken and estimated time to bring inflation back to the target.

Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the policy interest rate to signal the monetary policy stance.

Evaluation of Economic Conditions and Forecasts

The Bank of Thailand takes into account information from all sources, the macroeconomic model, data from each economic sector, as well as surveys of large enterprises, together with small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and cover all aspects, both at the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of monetary policy communication to the public, the MPC employs various channels of communication, both in Thai and English, such as (1) organizing a press statement at 14:00 on the day of the Committee meeting, (2) publishing edited minutes of the MPC meeting two weeks after the meeting, and (3) publishing the *Monetary Policy Report* every quarter.

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Growth and Inflation Prospects and Monetary Policy

1. Growth and Inflation Prospects and Monetary Policy

The Thai economy is projected to grow in 2016 at a pace slower than estimated in the previous *Monetary Policy Report*. Exports are restrained by both the slowdown in the global economy and structural problems in Thailand's export sector. Private consumption and private investment will consequently moderate. Meanwhile, growth in tourism and the government's accelerated disbursement of funds and various economic stimulus measures will only partially compensate for the weakness in economic momentum. For 2017, the Thai economy is projected to recover gradually, with domestic demand acting as the main driver and tourism providing continued support.

Headline inflation forecast for 2016 is revised down from the previous projection owing to a significant decline in crude oil prices. Core inflation forecast remains close to the previous projection, because a hike in the excise tax for tobacco products partly offsets the impacts from weakening domestic demand. For 2017, the Committee projects headline inflation to steadily rise in line with a gradual increase of global crude prices. Core inflation is expected to edge up slowly as well, but the forecast is subject to downside risks if demand pressures turn out to be less than expected.

The Committee kept the policy interest rate unchanged at the February and March 2016 meetings. Financial conditions remain accommodative, as reflected in real interest rates being negative and corporate funding costs remaining low. Nevertheless, the Committee is mindful of risks to financial stability, especially the search for yield behavior and a possibility that investors and financial institutions may have underpriced risks, which could contribute to a buildup in vulnerability in the domestic financial system. In addition, the Committee agrees that monetary policy divergence in advanced economies has contributed to capital flow volatility and baht appreciation in some recent episodes, which might not be conducive to the economic recovery as it could be. The Committee will monitor the situation closely.

1.1 Growth and inflation prospects

The Committee revises down the growth forecast for 2016 from 3.5 percent to 3.1 percent (Table 1.1) following the weaker outlook for merchandise exports on account of the slowdown of trading partners' economies, in particular Asian economies, and also the moderation in global trade due to shifts in the global trade structure. Moreover, declining export prices in line with global commodity prices weigh on exporters' incomes. This combination of a decline in the volume of exports and weak agricultural prices in line with crude prices has given rise to concerns for households about future incomes. Consequently, despite lower costs of living on the back of a decline in domestic energy prices, households become more cautious in spending, leading to private consumption slowing more than previously assessed. Furthermore, weakening demand from abroad and the slowdown in domestic consumer spending have caused firms to put on hold plans for expanding production capacity. Consequently, private investment is likely to grow at a lower rate than previously estimated, despite the anticipated increase in telecommunications investment expected to begin in the first quarter of 2016 after the completion of the auction for licenses to operate the 1800 MHz and 900 MHz frequencies.

The public sector has contributed to supporting the economy more than previously projected, as reflected in increased public investment thanks to greater-than-expected efficiency in disbursing funds, together with an increase in the budget for fiscal 2016. The government has earmarked the majority of the revenues to be collected from last year's auction

Table 1.1 Forecast summary

Percent	2015*	2016	2017
GDP growth	2.8	3.1	3.3
		(3.5)	
Headline inflation	-0.9	0.6	2.2
		(0.8)	
Core inflation	1.1	0.8	1.1
		(0.9)	

Note : *Outturn
( ) December 2015 MPR
Source : Office of National Economic and Social Development Board, Ministry of Commerce; calculations by Bank of Thailand

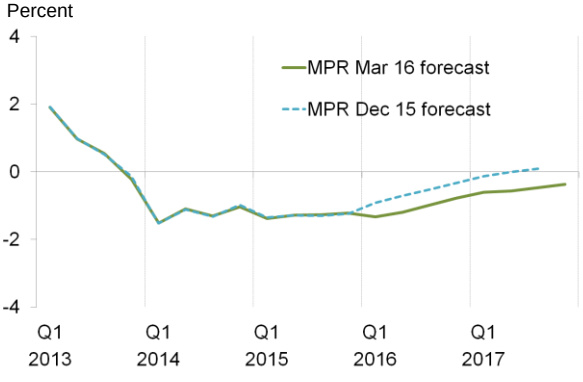
of 4G licenses for developing information technology infrastructure. Moreover, the government has launched additional stimulus measures and expanded the budget for existing stimulus measures, which should lend support to private spending to some degree.

Tourism is projected to grow more than previously estimated on account of a rising number of tourists especially from China. More Chinese tourists continue to visit Thailand, reflected growing popularity of the country as a tourist destination. Meanwhile, the number of tourists from other countries is likely to gradually recover, supported by more low-cost airline services to Thailand.

For 2017, the Thai economy is projected to recover gradually with a growth rate of 3.3 percent. Domestic demand will be the main driver, supported by tourism which continues to grow strongly. The export sector will still be weighed down by structural problems, however, causing private investment to expand only at a subdued rate **(Table 1.1)**.

With inflationary pressures turning out weaker due to both lower cost-push and demand-pull pressures than anticipated in the previous *Monetary Policy Report*, the Committee therefore revises down the projection for headline and core inflation for 2016 to 0.6 percent and 0.8 percent respectively. The main reasons are (1) domestic energy prices have recently declined in line with crude oil prices; (2) crude prices are expected to rise more slowly going forward than previously estimated; and (3) demand-pull price pressures have declined further given that the output gap is expected to close later than previously projected **(Chart 1.1)**. However, the hike in the tobacco

Chart 1.1 Output gap



excise duty in the first quarter of 2016 partly helps push up inflation, resulting in readings on core inflation edging down only slightly. **For 2017, the Committee projects inflation to be higher than 2016 on account of the increase in oil prices and the narrowing of the output gap. Headline inflation is forecast to be 2.2 percent and core inflation 1.1 percent (Table 1.1).**

The Committee has incorporated key economic developments into the growth and inflation forecasts as summarized below.

(1) Trading partners' economies are likely to grow at a slower pace than previously assessed (Table 1.2).

The Committee revises down the projection of trading partners' GDP growth throughout the forecast horizon. The downward revision is due to a further slowdown in Asian countries' exports and a weaker recovery in advanced economies than previously assessed. The latter weighs directly on demand for Thai exports and also indirectly on export prices, as the recovery in commodity prices will be slower than expected.

The sluggish global recovery has led to further monetary easing by advanced economies. The Committee anticipates that the Federal Reserve will raise its policy interest rate only twice this year, a more gradual pace compared with four rate hikes assessed previously. This is due to (1) the slowdown in Asian economies has adversely affected the U.S. recovery more than previously projected and (2) inflation likely remains low because of lower oil prices and a stronger U.S. dollar. Meanwhile, the Bank of Japan and the European Central Bank have eased their monetary policy further in order to support economic recovery. The Committee therefore

Table 1.2 Growth assumptions for Thailand's trading partners

Percent (%YoY)	Weight (%)	2015	2016		2017
			Dec 15	Mar 16	Mar 16
United States	14.9	2.4	2.6	2.2	2.4
Euro area	10.0	0.9	1.4	1.4	1.6
Japan	13.6	0.4	1.1	0.8	0.4
China	15.7	7.3	6.5	6.5	6.4
Asia (ex Japan and China)*	37.4	3.5	3.9	3.5	3.9
Total**	100	3.2	3.4	3.1	3.3

Notes: * Weighted by each trading partner's share in Thailand's total exports in 2014, namely Singapore (6.5%), Hong Kong (7.9%), Malaysia (8%), Taiwan (2.5%), Indonesia (5.9%), Korea (2.8%), and the Philippines (3.7%))

** Weighted by each trading partner's share in Thailand's total exports

expects that this additional monetary policy easing by advanced economies will contribute to a return of capital flows to Asian emerging markets to seek the higher relative returns, leading to increased volatility in capital flows going forward.

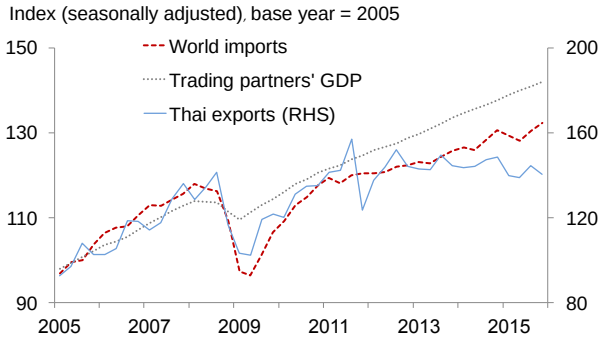
The Committee judges the balance of risks to trading partners' growth to be on the downside due to the following factors. First, Chinese and Asian economies may slow further, causing negative impacts on advanced economies. Second, the European banking problem may damage investor confidence and weigh on the economic recovery more than assessed. Third, geopolitical risks may prolong and affect trading partners' economies more than previously assumed. On the other hand, trading partners' economies may expand more than the baseline assessment if the additional monetary easing turns out to have supported advanced economies' recovery more than expected.

(2) Structural problems have affected merchandise exports more than previously assessed.

In addition to the slower-than-expected trading partners' recovery and weak commodity prices, Thai exports have been restrained by both external and domestic structural problems. External structural problems come from shifts in the global trade pattern such that several countries increasingly rely less on imports and more on intraregional trade, especially countries in Europe and Latin America. Moreover, Thailand's manufacturing sector has faced a number of structural problems that result in losses of competitiveness in global markets. First, the severe flooding in 2011 and political uncertainty have made Thailand less successful in attracting foreign direct investment. Second, changes in global consumer preferences have resulted in the

substitution of personal computers with smartphones and tablets. Consequently, as a major world producer of hard disk drives that are components for personal computers, Thailand has been significantly affected. Third, the loss of the European Union's Generalized Scheme of Preferences (GSP) privileges makes Thai exports less competitive compared with countries with the GSP. These structural problems have resulted in Thailand benefiting less from the recovery of trading partners' economies **(Chart 1.2)** (*Box: Impacts of Cyclical and Structural Factors on the Demand for Thai Exports*). **The Committee therefore projects a delay in the recovery of Thai exports and revises down the projection of this year's export value to be a contraction of 2 percent from last year, attributable to a decline in both quantity and prices.**

Chart 1.2 World imports, trading partners' GDP, and Thai exports



Source: Netherlands Bureau for Economics, calculation by Bank of Thailand

This weaker outlook for exports weighs on domestic spending and investment. Households have become more cautious in spending due to concerns about employment and income prospects, especially those in connection with the export-oriented manufacturing industries. This has led to a slowdown in private consumption, especially evident in the aftermath of a surge in consumption, which can be attributed partly to accelerated automobile purchases prior to the excise tax increase and partly to the government's economic stimulus measures at the end 2015.[1/] Moreover, falling global and domestic demand affects firms' income prospects, and together with existing spare capacity firms have not seen the need for investment to expand production capacity. Thus, the recovery in private

[1/] The government's tax deduction measure at the end of 2015 allowed taxpayers to claim an income tax deduction by the amount spent on goods and services during December 25-31 up to 15,000 baht.

investment is expected to take place later than previously estimated.

(3) The number of tourists has risen significantly and will likely increase more than the previous assessment.

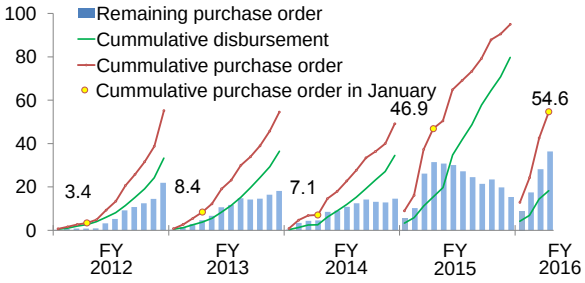
Exports of services are expected to grow strongly, as reflected in the increase in the number of tourists that is higher than previously forecast. This is a result of more low-cost airline routes to Thailand due to intensifying competition. Moreover, more Chinese tourists travel by cars crossing the northern border to Thailand via new routes. The number of Chinese tourists has thus grown rapidly, accounting for the highest share of foreign visitors to Thailand in 2015 and expected to continue growing in 2016 and 2017. In addition, the number of tourists from other countries likely continues to recover. The Committee therefore expects the number of tourists to reach 32.4 million in 2016, up from the previous forecast of 30.9 million, and 34.4 million in 2017.

(4) An increase in public investment that is more than expected and additional economic stimulus measures have contributed to sustaining the economy.

The Committee revises up the projection of public investment from the previous assessment in accordance with the government's investment budget spending with details as follows. **First, the government's measures to increase efficiency in spending have accelerated investment commitment and disbursement of funds** at the beginning of the fiscal year **(Chart 1.3)**. Consequently, this year's disbursement of the central government's investment budget is expected to be higher than the previous estimate. **Second,**

Chart 1.3 Central government's purchase order and disbursement of capital budget

Percentage of total capital budget



Note: Excluding transfers to local government and revolving fund

Source: Comptroller General's Department

the National Legislative Assembly has approved a supplementary budget for fiscal 2016,[2/] given an increase in revenues from the auction of licenses to operate 1800 MHz and 900 MHz, of which the government has earmarked the majority for developing information technology infrastructure. **Third, the Cabinet has approved a draft budget for fiscal 2017**[3/] **that is larger than expected, especially investment expenditure.** Incidentally, state-owned enterprise investment has been in line with the previous projection; extra-budgetary investment is slightly delayed, attributed to investment in water resource management projects by the Royal Irrigation Department.

In addition to an increase in investment spending, the government has launched additional economic stimulus measures in the first quarter of 2016[4/] and expanded the budget for existing measures,[5/] which should lend support to private consumption and investment to some extent. Furthermore, government consumption has turned out to be close to the Committee's previous projection.

The Committee judges the balance of risks to public spending to be titled to the upside. Upside risks involve the possibility that the government may speed up disbursement of on-budget and extra-budgetary investment outlays more than expected. Furthermore, state-owned enterprises may be able to expedite new investment sooner than anticipated.

[2/] Approved on February 18, 2016.

[3/] Approved on February 2, 2016.

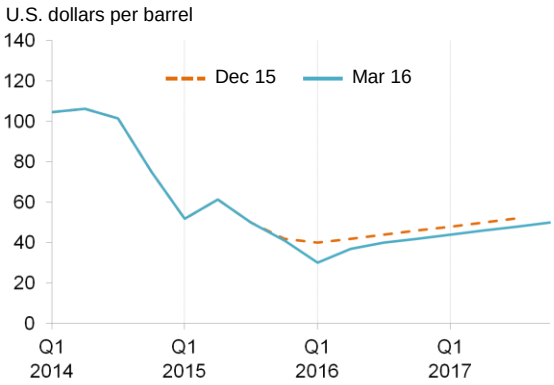
[4/] Stimulus measures aimed at strengthening the economy at the grassroots level in line with *Pracha Rat*: state funds will be provided to Village Funds to support infrastructure investment projects in local communities.

[5/] Stimulus measures providing low interest rate loans to SMEs and for property purchases.

In contrast, downside risks are related to limited efficiency of spending disbursement by government agencies.

(5) Crude oil prices drop lower than expected in the first quarter of 2016 and are projected to recover at a pace slower than previously estimated over the forecast horizon (Chart 1.4).

Chart 1.4 Assumptions on Dubai oil price



The Committee revises down the projection of the Dubai crude price throughout the forecast horizon. The oil price is expected to average at 37.3 U.S. dollars per barrel in 2016, lower than the previous estimate of 43.0 dollars per barrel. This downward revision is due to the more-than-expected decline in prices in the first quarter of 2016 because of concerns about elevated excess supply and warmer-than-usual weather in the U.S. and Europe that lowered demand for heating energy. Going forward, crude oil prices are projected to gradually increase on account of the reduction in supplies from Non-OPEC producers, especially from shale oil producers whose investment in exploration and production has been put on hold or canceled.

Implications for the global economy

Subdued crude oil prices should help support global recovery to some degree through lower production costs. However, the decline in oil prices negatively affected the recovery of oil-producing countries because of export revenue shortfalls and delayed investment spending by energy companies. Thus, the benefits to the global recovery from subdued crude prices are limited to some extent.

Implications for the Thai economy

Crude prices are expected to increase more gradually than previously expected, leading

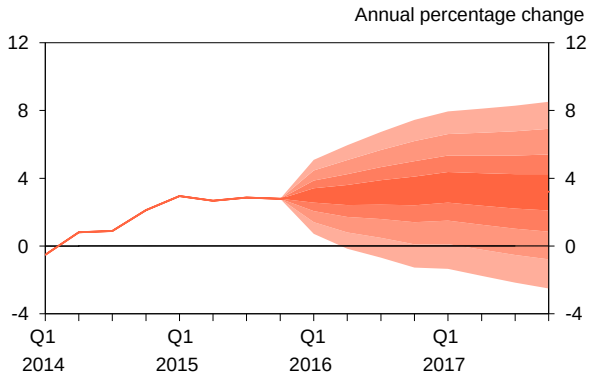
to a downward revision of domestic retail oil prices and headline inflation. Nevertheless, the Committee projects headline inflation to turn positive in the first half of 2016 following the waning of the base effect. In addition, subdued crude prices weigh on prices of Thai exports and agricultural products that are related to commodity prices. This further affects business revenues and farm incomes. Although the decline in crude prices helps reduce production costs of businesses and supports spending power of households through lower costs of living, it does not fully offset the reduction in export revenues and farm incomes.

The Committee judges the balance of risks to crude oil prices to be balanced. **Downside risks** are as follows. First, global oil demand may be lower than expected given a delayed global recovery. Second, despite lower crude prices, shale oil supplies may fall at a slower rate than anticipated. Third, OPEC producers may increase crude supplies more than the baseline projection in order to maintain their market shares. On the other hand, **upside risks** are as follows. First, OPEC producers and Russia may successfully cooperate to limit crude production. Second, the Middle East conflicts may spread to major oil production locations.

Risks to Growth and Inflation Forecasts

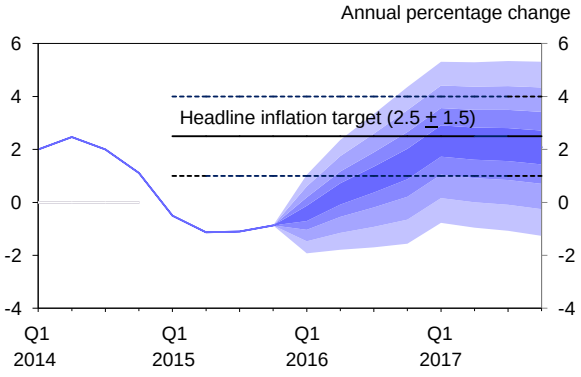
The Committee continues to judge risks to growth to be skewed to the downside but have become more balanced than previously assessed, as reflected in the growth fan chart that is less skewed throughout the forecast horizon (**Chart 1.5**). **Downside risks** to growth are as follows. First, the slowdown in Chinese and Asian economies may weigh on advanced economies more than the baseline assessment. Second, structural problems

Chart 1.5 GDP growth forecast



Note: Fan chart covers 90 percent of probability distribution

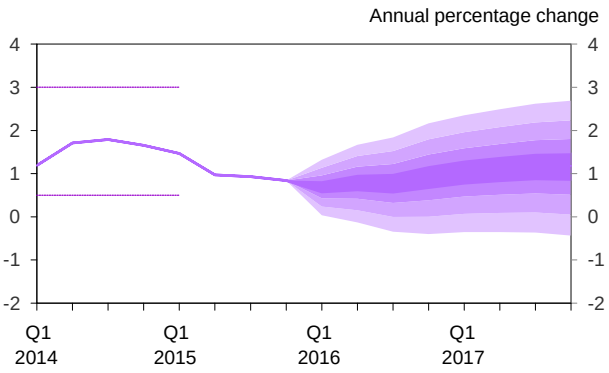
Chart 1.6 Headline inflation forecast



Note: Fan chart covers 90 percent of probability distribution

may cause Thai exports to benefit less from trading partner's economic recovery than anticipated in the baseline. On the contrary, the **upside risks** are as follows. First, fiscal expenditure, especially investment spending, may be executed sooner and greater than expected in the baseline. Second, the number of tourists may be greater than the baseline assessment. **Finally, the Committee judges the balance of risks for both headline and core inflation to be tilted to the downside in line with the balance of risks to growth (Chart 1.6 and 1.7).**

Chart 1.7 Core inflation forecast



Note: Fan chart covers 90 percent of probability distribution

1.2 Monetary policy decision

Monetary policy remains accommodative in order to facilitate economic recovery in the face of more adverse risks, especially external risks, during the first quarter of 2016. Meanwhile, inflationary pressures have declined following decreases in global oil prices. Given the high level of uncertainty in connection to the global economy, the Committee sees the need to preserve policy space. In addition, risks to financial stability from the search for yield behavior, of which signs are increasingly observed, continue to warrant close monitoring.

Nevertheless, monetary policy divergence among advanced economies has caused the baht to strengthen during some recent periods, which might not be as conducive to the economic recovery as it could be. The Committee will thus monitor the situation closely, continue to maintain the accommodative policy stance to support Thailand's economic recovery, and stand ready to utilize an appropriate mix of available policy tools, together

with being mindful of imbalances which may build up in a prolonged low interest environment.

At the MPC meeting on February 3, 2016 the Committee voted unanimously to maintain the policy rate at 1.50 percent. Under the Committee's assessment, the economic and inflation outlook did not change significantly from assessment in the previous meeting. The Thai economy and inflation were expected to gradually recover, albeit with increasing downside risks to growth that were largely attributable to external factors, in particular the global economic slowdown and subdued commodity prices. The Committee noted that monetary policy remained accommodative, as reflected in a negative real policy rate[6/] and the low financing costs of businesses relative to the past average level, which should be conducive to fostering economic recovery and returning inflation gradually to the target.

Some members of the Committee expressed a view that further monetary policy easing could only marginally improve the economic and inflation outlook. The levels of interest rates were deemed not a factor that held up business operations or household consumption under the current policy stance that was already accommodative. In addition, some members were concerned about the appreciation of the Nominal Effective Exchange Rate (NEER) that might not be as conducive to Thailand's economic recovery as it had been in the past. Consequently, the

[6/] The real interest rate is calculated by subtracting one-year-ahead inflation expectations, assumed to equal the one-year-ahead forecast of headline inflation (about 2.3 percent according to Chart 1.6), from the policy interest rate (1.5 percent as of the first quarter of 2016). The real interest rate is thus approximately -0.8 percent in the first quarter of 2016.

Committee would pay close attention to global economic and financial developments, particularly monetary policy divergence among advanced economies.

The MPC concurred that monetary policy at this juncture should be attentive to risks to financial stability, in particular the search for yield behavior, of which signs could be increasingly observed in the prolonged low interest rate environment. The Committee also saw the need to preserve policy space for future use, as there remained a high degree of uncertainty especially in the external front, for instance China's economic and financial developments, from which adverse spillovers could affect Thailand's economic recovery more than anticipated.

At the meeting on March 23, 2016 the Committee projected the economy to grow at a lower rate than previously anticipated. The downward revision was partly on account of weaker private consumption due to the dissipation of the temporary boosts from the tax deduction measure at the end of last year and the accelerated automobile purchases prior to the vehicle excise tax hike. Furthermore, the value of merchandise exports still contracted significantly and looked to rebound only slowly as a result of major trading partners' economic slowdown, a decline in Thailand's export competitiveness, and the shifts in the global trade structure that led advanced economies to import less. Meanwhile, inflation pressures remained low, as reflected in headline inflation remaining negative and core inflation, with the impacts of the increases in the excise taxes on tobacco products and automobiles excluded, edging lower in line with weakening price pressures from the demand side.

Accordingly, the Committee deemed risks to inflation skewed to the downside.

Nevertheless, with monetary policy still being accommodative, the MPC concurred that it should preserve policy space for future need and that at the current juncture the policy rate might not be the most effective tool to boost the economy. Under these circumstances, the Committee voted unanimously to leave the policy rate unchanged at 1.50 percent at this meeting. Meanwhile, some members were concerned about the impact of monetary policy divergence on the extent of the baht appreciation that was relatively more than most regional currencies during some recent periods, which might not be as conducive to the economic recovery as it could be. The Committee would therefore monitor global economic and financial market developments closely.

The Committee assessed overall financial stability risks to be increasing somewhat on account of the search for yield behavior, together with a deterioration in household income and commercial banks' asset quality due to the delayed economic recovery. In this prolonged low interest rate environment, the Committee would also monitor signs of underpricing of risks by investors and financial institutions that could lead to a buildup of vulnerability in the Thai financial system.

Looking ahead, the Committee concurred that monetary policy should continue to remain sufficiently accommodative. The Committee would stand ready to utilize an appropriate mix of available policy tools in order to support the economic recovery, bring inflation rate back to the target within a reasonable time period, and ensure financial stability.

1.3 Appendix: Tables for supporting assumptions and forecasts

Table 1.3 Forecasts for GDP and components

Percent	2015*	2016	2017
GDP growth	2.8	3.1	3.3
Domestic demand	2.8	2.8	2.9
Private consumption	2.1	1.8	2.4
Private investment	-2.0	2.4	4.0
Government consumption	2.2	3.3	2.8
Public investment	29.8	10.7	4.5
Exports of goods and services	0.1	1.0	1.7
Imports of goods and services	-0.4	0.6	1.5
Current account (billion, US dollars)	34.8	34.8	32.7
Value of merchandise exports	-5.6	-2.0	0.1
Value of merchandise imports	-11.3	-6.1	3.6

Note: *Outturns

Table 1.4 Forecast assumptions

Annual percentage change	2015*	2016	2017
Dubai oil price (U.S. dollars per barrel)	51.0	37.3	47.0
Non-fuel commodity prices (%YoY)	-17.5	-6.9	2.7
Fresh food prices (%YoY)	-4.4	3.0	4.0
Monimum wage in the Bangkok Metropolitan Region (baht per day)	300	300	300
Government consumption (current price) (%YoY) 1/	4.4	6.1	6.4
Public investment (current price) (%YoY) 1/	25.6	10.8	7.9
Fed Funds rate (% at year-end)	0.38	0.88	1.38
Trading partners' economic growth (%YoY) 2/	3.2	3.1	3.3
Regional currencies vis-a-vis the U.S. dollar (Index) 3/	150.7	160.6	161.4

Note : 1/ Including spending on water management plans and infrastructure investment projects
2/ Weighted by each trading partner's share in Thailand's total exports
3/ Appreciation against the US dollar indicated by the minus sign
* Outturns

Table 1.5 GDP growth forecasts by research houses

	2016	2017
Standard Chartered	4.0	5.0
Barclays	3.7	4.0
FPO[1/]	3.7	-
ANZ Bank	3.5	3.8
DBS	3.4	3.7
NESDB[2/]	3.3	-
Phatra	3.2	3.6
Kiatnakin	3.2	3.6
Bank of Ayudhya	3.2	3.5
BOT	3.1	3.3
Moody's	3.0	3.3
Kasikorn Research	3.0	3.2
JP Morgan	2.9	3.0
Credit Suisse	2.7	3.1
SCB	2.5	3.3
Nomura	2.5	2.7

Notes: Compiled and published by Reuters on March 23, 2016, except:
[1/] Published on January 28, 2016
[2/] Published on February 15, 2016, with the release of GDP data for 2015 Q4
Presented in descending order of 2016 forecasts

Table 1.6 Headline inflation forecasts by research houses

	2016	2017
Moody's	2.3	2.9
Standard Chartered	1.4	2.3
SCB	0.8	2.9
BOT	0.6	2.2
DBS	0.6	2.0
ANZ Bank	0.6	2.0
Bank of Ayudhya	0.6	1.5
Credit Suisse	0.5	2.0
Barclays	0.5	2.0
Kiatnakin	0.5	1.8
Phatra	0.5	1.8
Citi	0.4	2.2
Kasikorn Research	0.4	1.7
NESDB[1/]	0.4	-
Nomura	0.3	1.0
FPO[2/]	0.3	-
KT Zmico	0.0	1.5

Notes: Compiled and published by Reuters on March 23, 2016, except:
[1/] Published on January 28, 2016
[2/] Published on February 15, 2016, with the release of GDP data for 2015 Q4
Presented in descending order of 2016 forecasts

Table 1.7 Probability distribution of GDP growth forecast								
Percent	2016				2017			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
10-12	0	0	0	0	1	1	1	1
8-10	0	0	1	3	4	4	4	5
6-8	2	8	13	17	18	16	16	16
4-6	18	21	21	20	19	17	16	16
2-4	54	40	34	29	27	26	24	23
0-2	24	23	22	20	19	20	20	19
(-2)-0	2	5	7	9	9	11	11	12
< (-2)	0	1	2	3	3	5	6	8

Table 1.8 Probability distribution of headline inflation forecast								
Percent	2016				2017			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
6-7	0	0	0	0	2	2	2	2
5-6	0	0	0	2	5	5	5	4
4-5	0	0	1	5	10	10	10	9
3-4	0	1	5	11	17	17	16	15
2-3	0	6	14	18	21	21	20	19
1-2	5	19	23	22	20	19	19	19
0-1	27	30	25	20	14	14	14	15
(-1)-(0)	42	27	18	13	8	8	8	9
(-2)-(-1)	22	13	9	6	3	3	4	4
(-3)-(-2)	4	4	3	2	1	1	1	2
(-4)-(-3)	0	1	1	1	0	0	0	0

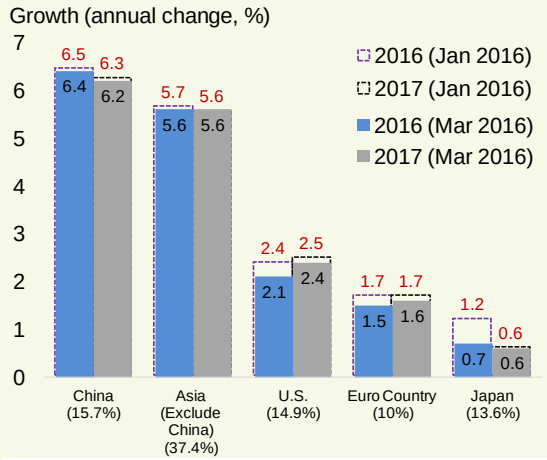
Table 1.9 Probability distribution of core inflation forecast								
Percent	2016				2017			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
3.0-3.5	0	0	0	0	1	1	1	2
2.5-3.0	0	0	0	1	3	3	4	5
2.0-2.5	0	1	3	6	8	9	10	11
1.5-2.0	1	8	11	14	17	17	18	17
1.0-1.5	18	26	24	23	23	22	22	21
0.5-1.0	47	35	29	24	22	21	20	19
0.0-0.5	29	22	21	18	15	14	14	13
(-1)-0.0	5	7	9	9	8	7	7	7
(-2)-(-1)	0	1	3	3	3	3	3	3
< -2	0	0	1	1	1	1	1	1

Impacts of cyclical and structural factors on the demand for Thai exports

The fact that export volumes of Asian countries and Thailand experienced a contraction in 2015, despite global trade volumes and the trading partners' economies still expanding albeit at a slowing rate, points to significant adverse effects of structural factors on exports of many Asian countries including Thailand. Looking ahead, exports of Asian countries, including Thailand, may not record high growth rates as in the past despite a recovery in advanced economies.

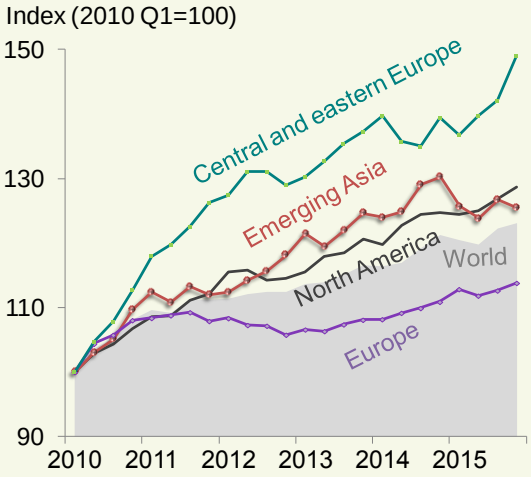
The contraction in exports is witnessed in several Asian economies. This is not only due to a cyclical slowdown in China and a slower-than-expected recovery of major economies **(Chart 1)** but also due to structural factors. The impact of structural factors on exports is estimated to be increasingly pronounced and appears to weigh on exports more than cyclical factors do recently. This is reflected in a fall in export volumes of Asian countries in 2015 despite expansion in global trade volumes, particularly in North America and Europe **(Chart 2).**

Chart 1 Growth assumptions for Thailand's trading partners



Note: Shares in total Thai exports are shown in brackets
Source: Consensus Forecast

Chart 2 Trade volume by region

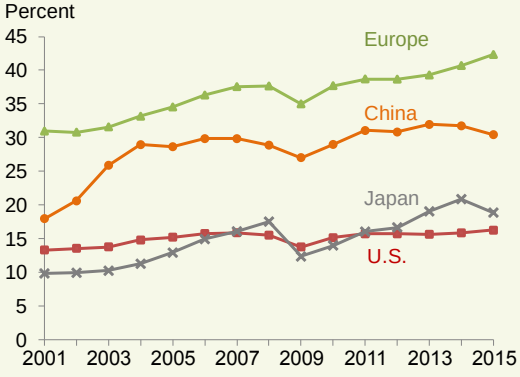


Source: CPB Netherlands Bureau for Economic Policy Analysis

The decline in regional trade volumes can be attributed to the following factors.

1) Changes in the global trade structure. Trading activities among Asian countries have become interconnected over the past decade. Several Asian countries are parts of the global value chain, whereby China imports intermediate goods from its regional trading partners for assembling and then exports as final products—in particular, electronics. However, China's ongoing economic reform affects the global value

Chart 3 Shares of imports to GDP

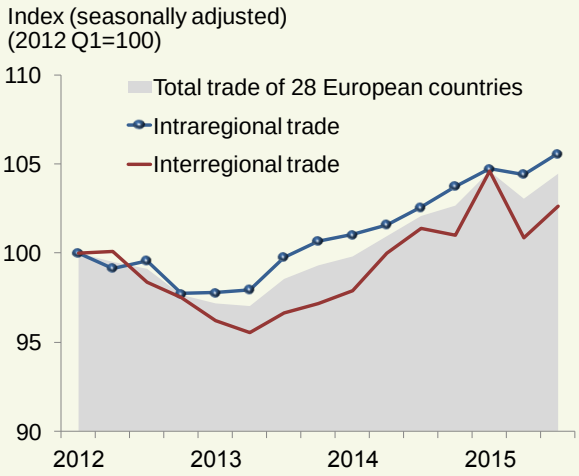


Sources: Oxford Economics and CEIC

chain in two ways. First, the reform aims at substituting exports and investment with domestic consumption and services as the growth engine. Second, China focuses on moving up the value chain and therefore effectively shortens the Asian supply chain. Both factors result in China importing less, particularly raw materials and intermediate goods, as reflected in a share of China's imports to GDP that starts to show a downward trend **(Chart 3)**.

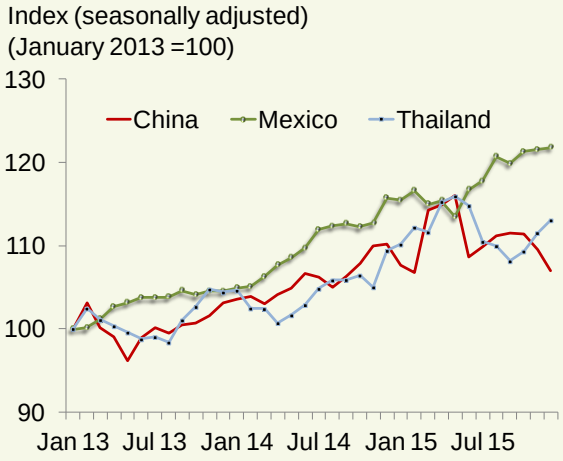
Moreover, the recent recovery of advanced economies appears to have benefited Asian exports less than in the past, as the U.S. and Europe have increasingly engaged in more intraregional trade[1] **(Charts 4 and 5)**. Given that Asia gradually loses cost competitiveness, multinational corporations start to shift their production bases back home or to countries which have relatively cheaper labor costs such as Mexico and Eastern European countries.

Chart 4 Europe's intraregional trade and interregional trade



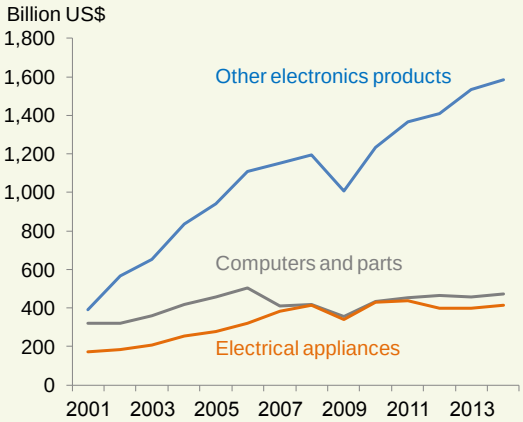
Source: UNCTAD

Chart 5 Non-energy imports by U.S. from selected countries



Source: CEIC

Chart 6 Global electronics exports



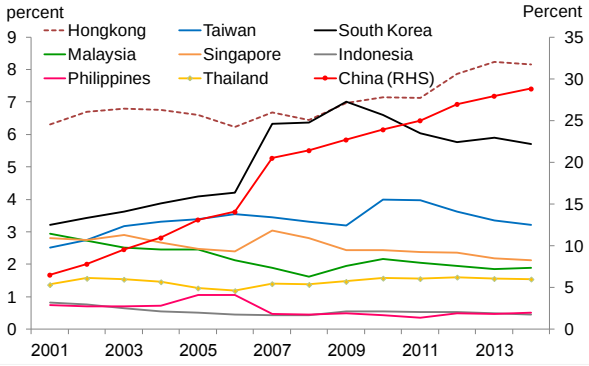
Note: Smartphones and tablets are in "Other electronics products"' category

Source: Trademap; calculation by Bank of Thailand

2) A shift in consumer preferences from computers to smart phones and tablets leads to a continuing rise in exports of smartphones and related electronic parts, while exports of computers and computer parts remain flat **(Chart 6)**. The impact of this change in preferences is not uniform across Asian countries. Manufacturers of computers and hard disk drives, as well as those in the supply chain, such as Thailand, are more adversely affected. In contrast, smartphone and tablet producers and those in the supply chain, such as Korea, Taiwan, and Vietnam, have benefited from this change in consumer

[1/] The importance of the global value chain in international trade has declined somewhat, as China advances its production technology, obviating the need to import from Asia, and as the U.S. and the European Union increasingly depend on regional value chains, making Asia not fully benefit from the economic recovery of advanced economies.

Chart 7 Market shares of high-tech products



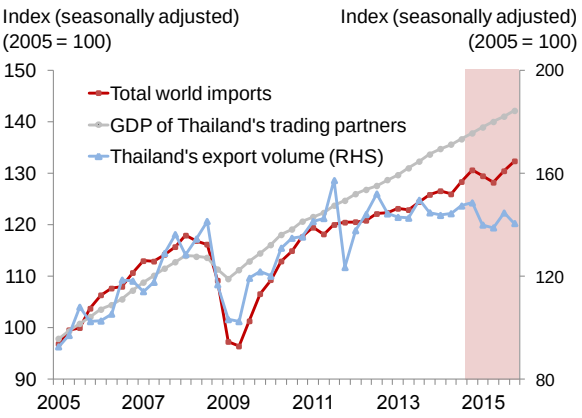
Note: High-tech products consist of communication devices, electrical appliances, photographic and cinematographic instruments, and optical appliances.
Source: Trademap

preferences, as reflected in strong growth in their electronics exports.

3) Intra-Asia trade has been trending down due either to slower domestic growth or to structural limitations. First, the loss of competitiveness to China has caused intraregional trade in Asia to slow down. Over the years, China has successfully shifted from assembling parts for multinationals in the U.S. or Korea to establishing its own brands, which offer similar quality but at lower prices. As a result, China's market share for high-tech products has been rising constantly, while those of other countries, such as Taiwan and Korea, have been on a downward trend **(Chart 7)**, especially in the category of high-tech products such as smartphones. Second, aging population has become more critical in several Asian economies. This results in a shrinking labor force and therefore a lower growth momentum contributed by labor. In addition, aging population increases the likelihood of the middle income trap, especially if the economy fails to have a new growth driver or attract foreign investment as previously. In sum, these structural limitations have adversely affected intraregional trade in Asia.

Thailand has its own set of domestic structural challenges on top of the above global and regional structural changes. In particular, the loss of export competitiveness is reflected in export contraction for three consecutive years **(Chart 8)**, and the problem becomes increasingly pronounced. Thailand's structural challenges can be summarized as follows.

Chart 8 Global trade volume and Thailand's trading partners' GDP



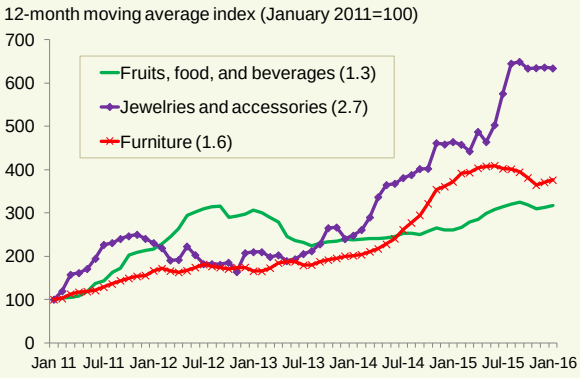
Source: Thailand's Customs Department, Ministry of Commerce, Bank of Thailand, and CPB Netherlands Bureau for Economic Policy Analysis

1) Due to a persistent lack of both public and private investment, Thailand has not been able to secure technological advancement needed for keeping up with changes in consumer preferences. Private investment, for instance, has been subdued due to labor shortage, both in terms of quantity and quality. Moreover, a quite rapid rise in the minimum wages has made Thailand relatively less attractive as a destination for foreign investment compared with its regional peers. As a result, the level of foreign investment in Thailand's manufacturing industries has been trending downward. At the same time Thailand's direct investment abroad has constantly been on the rise. Furthermore, research and development (R&D) has received limited attention in Thailand, as reflected in (1) the share of R&D expenses to GDP, (2) R&D expenses per capita, and (3) the number of personnel engaged in R&D activities. Based on these indicators, Thailand's R&D performance is quite low compared with regional counterparts. On top of that, prolonged political instability

and limitations in fiscal disbursement have delayed infrastructure investment. Without the necessary adjustments on these fronts, the export sector will not be able to resume its role as Thailand's key growth driver as in the past.

2) The termination of special trade agreements further contributes to a loss of export competitiveness. From 2015 onward, all exports items from Thailand are no longer qualified for the Generalized Scheme of Preferences (GSP) from the European Union, which is among Thailand's top trading partners, with a share of 10.2 percent of total Thai exports in 2015. As a result, exports to the European Union dropped by as much as 6 percent, particularly textiles and apparels, processed food, and fisheries. Going forward, the competitiveness of Thai exports could face further challenges, as Thailand's Free Trade Agreement (FTA) negotiations have seen limited progress compared with other countries in ASEAN such as Vietnam, Malaysia, and Singapore. In the absence of tax incentives from such FTAs, there is little to help offset the adverse impacts of the GSP termination. Moreover, Thailand might become less attractive as a foreign investment destination compared with neighboring countries that have signed the Trans-Pacific Strategic Economic Partnership Agreement (TPP).

Chart 9 Selected merchandise exports to China



Note: Shares in total Thai exports to China are shown in brackets
Source: Thailand's Customs Department, and Bank of Thailand

Although the change in the global trade structure is beyond Thailand's control, it is possible for Thai businesses to adapt to the new landscape. For instance, thanks to a growing number of Chinese tourists in Thailand, Chinese people have become more familiar with Thai products. In particular, Thai consumer products meet the quality standards and are widely accepted by the Chinese. Products like fruits, food, beverages, jewelries and accessories, and furniture have therefore seen steady increase in exports to China **(Chart 9)**. (Exports of raw materials and intermediate goods such as rubber, rubber-related products, petrochemicals, and chemicals in contrast have been trending down, as China replaces its imported components in heavy industries with domestic production.) However, the exports of the consumer products to China that perform well account for only 5.6 percent of Thailand's total exports to China. Going forward, both the public and the private sectors should therefore attempt to promote exports of these products, with particular focus on adding value to existing products and maintaining their quality.

Regarding domestic structural limitations, both the public and the private sectors are aware of the issues and are in the process of implementing measures to raise the country's competitiveness as follows.

- Identify industries which have the potential to become a new growth engine. First, the government has already set a target for investment incentives for five industries which have the potential to boost growth in the short to medium term (First S-curve). This involves expanding their existing capability into more sophisticated technology, such as next-generation cars, smart electronics, medical tourism, agriculture and biotechnology, and food

processing. Simultaneously, another five industries will be promoted to serve as a longer-term growth engine (New S-curve), namely robotics for industry, aviation and logistics, biofuels and biochemical, digital, and medical sectors. Moreover, funds have already been established to help promote competitiveness of these industries.

- Promote research, technological advancement, and innovation that can also be applied for commercial use. In this regard, the government has set key milestones for the R&D target, namely 1 percent of GDP by 2016 and 2 percent of GDP by 2021, up from only 0.47 percent in 2013. Moreover, the government aims to raise the contribution of the private sector to national R&D expenses, from 47 percent in 2013 to 70 percent, and has introduced a number of measures to help gear its stakeholder toward the target. For instance, the tax deductible amount for R&D and innovation-related expenses are raised from 200 percent of the amount spent to 300 percent. Moreover, the Talent Mobility Program will supply the private sector with researchers from the public sector in order to ensure the commercial application of research. The Export-Import Bank of Thailand will also supply low-interest loans, while its counterpart, the National Science and Technology Development Agency, offers research grants for entrepreneurs.

- Accelerate progress on trade agreement negotiation. The government has already assigned this task to relevant departments, particularly on the issue of TPP membership. The final decision, however, is likely to be reached after thorough cost-benefit analysis has been undertaken. Given the time-consuming nature of the process, Thai exporters will have to wait for some time before receiving the tax preferences.

Based on slower-than-expected growth of trading partners' economies, more pronounced structural problems both at home and abroad, and the fact that the reform of Thailand's export sector is likely to take some time to materialize, the forecast of Thai exports in 2016 is thus revised down, with the likelihood that exports will contract for four consecutive years. Such contraction, however, is also seen in several Asian economies experiencing structural problems. For 2017, it is expected that exports growth will turn positive albeit at a low rate. This is because while export prices should pick up in line with a gradual increase in global commodity prices, the volume of exports is expected to continue shrinking as a result of the above structural problems.

While the export sector is undergoing reform, which might lead to the economy operating below potential in the short run, the government should focus its limited resources on strengthening growth momentum by investing to upgrade production efficiency and increase the country's competitiveness. This can be done by investing more in high value-added or innovative products, and promoting more concrete investment plans for the New S-curve industries. As a pilot project, the industries which have made the most progress should be selected in order to secure confidence of investors. Such investment should enhance the country's competitiveness in a sustainable manner compared with the consumption stimulus measures that result in only temporary boosts to the economy.

Recent Economic Developments

2. Recent Economic Developments

The global economy continued to show signs of recovery. Nevertheless, forward growth momentum was likely to moderate for both advanced and Asian economies. For advanced economies, the impacts from the slowdown in emerging market economies turned out greater than previously assessed primarily because of a drag in exports. However, domestic demand continued to be a main driver sustaining economic recovery going forward. For Asian economies, headwinds to growth came from both cyclical and structural problems. Cyclical problems were related to a delayed global recovery. Structural problems, which were attributable to a changing global trade structure and domestic structural constraints, weighed on growth more than previously anticipated.

The Thai economy showed signs of weakening in various sectors more evidently. The repercussions of the contraction in exports on overall economic activities were increasing. Private consumption slowed down after the effects of temporary boosts started to wane. Private investment, especially in the manufacturing sector, remained subdued, despite improvements seen in some services sectors such as telecommunications. Nonetheless, public investment and tourism continued to play a crucial role in sustaining the economy. Disbursement of government funds was on course, and there were indications of a rebound in the number of tourists from increasingly more countries, which would provide further support to the expansion in tourism-related services industries.

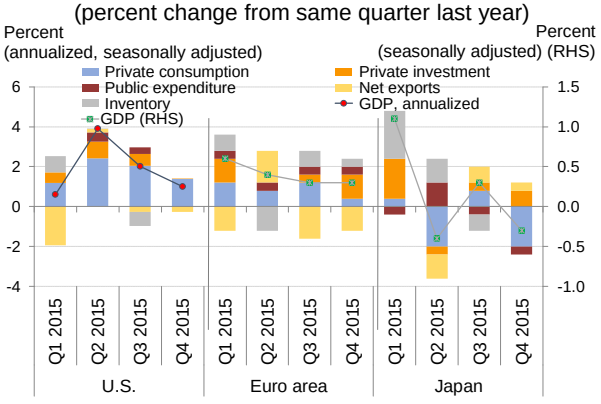
Regarding costs and prices, headline inflation remained negative given that global oil prices were still at low levels. Core inflation edged down on account of weaker domestic demand and subdued cost pressures.

2.1 The global economy

With regard to advanced economies, the drag from the slowdown in emerging market economies turned out greater than previously assessed owing to a decline in export activities **(Chart 2.1 and 2.2)**. However, domestic demand continued to be a main driver to support economic recovery going forward.

The U.S. economy continued to expand, albeit at a slower pace at 1 percent

Chart 2.1 Sources of GDP growth of G3 economies
(percent change from same quarter last year)



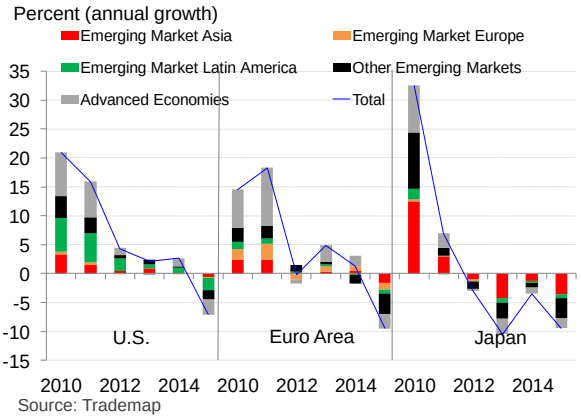
Source: Bureau of Economic Analysis, Eurostat, Cabinet Office of Japan

(qoq saar) in the fourth quarter of 2015,[1/] down from 2.0 percent in the previous quarter. Three factors contributed to the slowdown were (1) a significant moderation in investment following the contraction in investment in the energy sector that were adversely affected by persistently low oil prices; (2) a continued decline in inventories from elevated levels in the first half of the year; and (3) shrinking net exports due to weakening demand from abroad in combination with a stronger U.S. dollar.

Nonetheless, recent economic data indicated a continued recovery, driven primarily by domestic demand. In particular, private consumption was underpinned by rising incomes and continuing improvement in the labor market, as reflected in the improvement in retail sales. However, the speed of the U.S. recovery would likely be more sluggish than previously expected on the back of the slowdown in the global economy, especially emerging markets, which continued to weigh on U.S. exports. Given weakening exports and heightened volatility in the U.S. financial markets, the pace of the federal funds rate increases would likely be to more gradual than assessed in the previous *Monetary Policy Report*.

The euro area economies[2/] grew by 0.3 percent (qoq sa) in the fourth quarter of 2015, a rate unchanged from the previous quarter. Growth was primarily supported by continued

Chart 2.2 G3 exports classified by destination



Source: Trademap

1/ Economic data reported in this *Monetary Policy Report* were based on the official data which had been released up to March 22, 2016, the day before the March MPC meeting. The figure for the fourth quarter U.S. GDP growth reported above was the second estimate.

2/ Comprising 19 countries using the single currency

expansion in domestic demand, in particular private investment and public consumption. In contrast, private consumption moderated due to a dent in consumer confidence, in part because of the Paris terror attack in November 2015. Furthermore, latest data pointed to increased tightening in financial conditions following the recent European financial market volatility. Although this would further weigh on private consumption, both subdued oil prices and gains in employment could somewhat offset the negative impacts. Meanwhile, weak exports remained a drag on growth.

Going forward, despite the supports from an improving labor market, low oil prices, and accommodative monetary policy, a few adverse risks were present. One risk would be related to banking sector fragility, especially in the periphery with elevated nonperforming loans and financial sector links to emerging markets and oil industries. The other risk would be related to the negative interest rate policy adopted by the European Central Bank that could adversely affect profitability of commercial banks. These negative risk factors could create greater financial market volatility and tightening financial conditions that would partly restrain the support from monetary policy accommodation. Furthermore, uncertainties in connection with geopolitical risks and prolonged political tensions could disrupt the recovery of the euro area and the global economy. The Committee therefore would closely monitor these risks which could have implications for the conduct of monetary policy in Thailand going forward.

The Japanese economy contracted by 0.3 percent (qoq sa) in the fourth quarter of 2015, down from a 0.3 percent growth rate registered in the previous quarter due to a sharp drop in private consumption. The

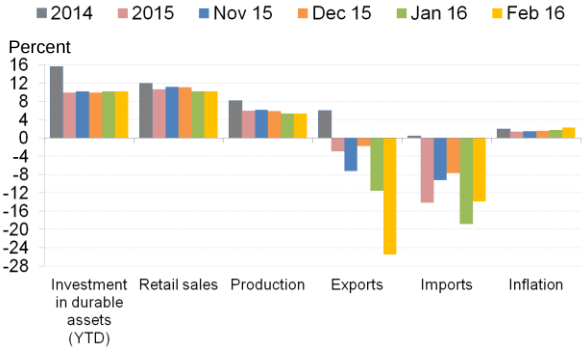
contraction in consumer spending was partly attributable to (1) temporary factors such as warmer-than-usual weather that caused a fall in winter clothing sales and energy consumption and (2) the elevated level of consumption in the previous quarter. Preliminary data for the first quarter of 2016 indicated a decline in consumer confidence and retail sales, which reflected adverse effects from volatility in the global financial market and declines in the prices of financial assets. Incidentally, private investment continued to expand in the fourth quarter.

Going forward, the Japanese economy would likely recover gradually, supported by a healthy labor market, household incomes, and strong corporate profits. However, notable risks to Japan were considered to be (1) a slowdown in external demand that could restrain exports and (2) a more-than-expected prolong of global financial market volatility that could hurt domestic demand through a decline in confidence.

The Chinese economy moderated on account of shrinking exports and weakened private consumption and also a slowdown in activities in the financial sector after the recent volatility. Consequently, the Chinese authority rolled out further stimulus measures to support the economy.

The Chinese economy expanded at a rate of 6.8 percent (yoy) in the fourth quarter of 2015, slightly down from 6.9 percent in the previous quarter on account of the moderation in the services industries that was attributed to the slowdown in financial sector activities. Nonetheless, latest indicators pointed to a further moderation in the economy **(Chart 2.3)** due to weakening consumption and shrinking exports, despite investment that expanded slightly thanks

Chart 2.3 China's economic indicators
(percent change from same month last year)



Source: CEIC

to monetary policy accommodation, real estate stimulus measures, and public investment. Meanwhile, manufacturing production would likely moderate because of falling external and domestic demand and policies to reduce excess capacity despite the continued expansion in services industries.

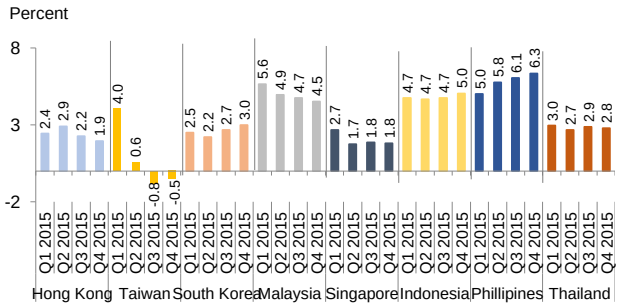
Given the prospect of an economic slowdown and liquidity tightening, the Chinese administration continued to announce stimulus measures to sustain the economy to achieve this year's growth target of 6.5-7.0 percent. These measures included an increase in the frequency of open market operations, a cut in the reserve requirement ratio, and real estate stimulus measures, namely an easing of mortgage down payment requirements and a reduction in transaction taxes for some home buyers.

Asian economies were adversely affected by both cyclical factors that delayed the global recovery and structural factors, namely a change in the global trade structure and domestic structural constraints, that weighed on economic growth more than previously expected.

Several **Asian economies (excluding China and Japan**) grew at a slower pace **(Chart 2.4)** due primarily to weakening domestic demand and greater contraction in exports **(Chart 2.5**). Exports of Asian economies were increasingly affected not only by the global economic slowdown, attributed especially to emerging market economies, but also by structural challenges related to both country-specific constraints and slower trade because of a change in the global trade structure.

Weakening exports adversely affected domestic demand in most Asian countries, as reflected in lower employment and incomes.

Chart 2.4 GDP growth of Asian economies
(change from same quarter last year)

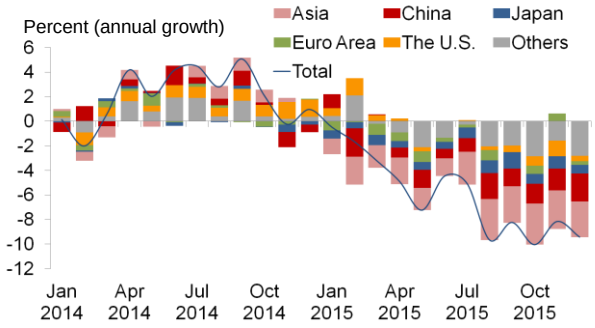


Source: CEIC

Latest data indicated that Asian economies would continue to slow as reflected in the Purchasing Managers Index (PMI) and its new export orders component of many countries remaining below 50.

Going forward, Asian economies (excluding China and Japan) would likely grow at a slower pace because of the softening in exports and domestic demand and because of domestic structural constraints such as heavy dependence on commodity exports, labor market problems, and population aging.

Chart 2.5 Contribution to Asia exports classified by destination



Note: Asia includes Hong Kong, Taiwan, South Korea, Malaysia, Singapore, the Philippines, and Thailand
Source: CEIC

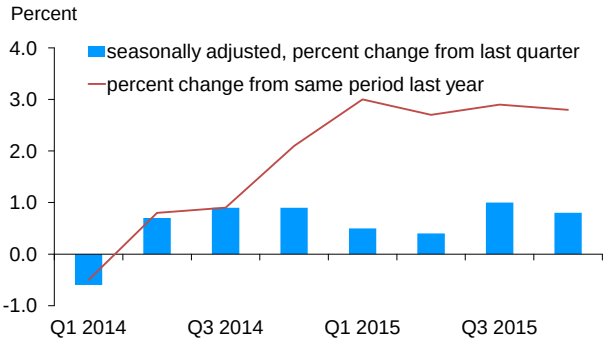
2.2 The domestic economy

Economic momentum showed visible signs of losing steam in several sectors, as export contraction started to increasingly impact the real economy while the positive effects from temporary factors started to wane. However, public investment and tourism remained key drivers in sustaining economic momentum (Chart 2.6).

Exports continued to contract in both quantity and value terms as cyclical and structural challenges increasingly weighed on Thai exports.

Adverse cyclical factors in the global economy led to greater negative impacts on exports of several countries, including Thai exports. Especially after the effects of temporary economic boosts began to fade, these negative impacts became more evident, as reflected in a slowdown in exports of some product categories, such as automobiles and electronic parts for new mobile phones and tablets which had previously accelerated following product launches. Overall, Thai exports to most destinations declined, except to Cambodia, Laos, Myanmar, and Vietnam

Chart 2.6 GDP growth[1/]
(seasonally adjusted, percent change from last quarter)



Note: [1/] Calculation based on chain volume measure (CVM)
Source: Office of the National Economic and Social Development Board

(CLMV) which continued to expand **(Chart 2.7).** In addition to affecting the volume of exports, the global economic slowdown continued to weigh on oil and other commodity prices. As a result, prices of exports associated with oil and commodities, for example, petroleum, petrochemicals, chemicals, and rubber and rubber products, contracted sharply.

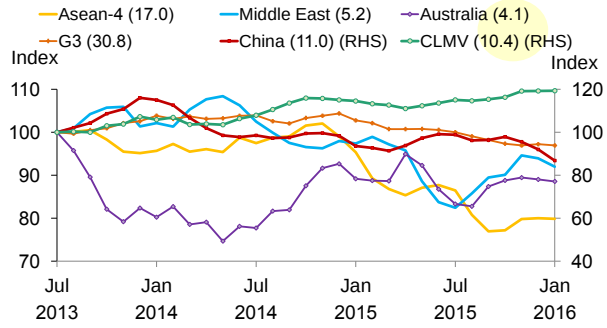
Apart from cyclical factors, structural problems increasingly weighed on Thai exports and would likely remain a drag for the foreseeable future. Structural problems could be classified into two categories as detailed below.

(1) Shifts in the global trade structure led to a decline in imports of goods and raw materials from abroad. First, the Chinese administration's emphasis on the importance of the domestic contents of exports in accordance with its economic reform policy increasingly affected other trading partners, especially ASEAN countries which were linked to China's production chain. Second, intraregional trade in the European Union was on the rise. Third, the U.S. now increasingly relied on imports from Mexico instead of China and Asia. As a result, these shifts in the global trade structure would continue to weigh on Thai exports.

(2) Domestic structural problems, such those related to labor and manufacturing technology, continued to limit competitiveness in production and exports. Examples included technological limitations in the production of solid state drives that were to replace hard disk drives, the latter of which faced a continuous decline in demand. Moreover, Thailand's loss of the Generalized Scheme of Preferences (GSP) privileges since the beginning of 2015 would affect

Chart 2.7 Thai exports by destination

(3-month moving average, July 2013 = 100)



Note: ( ) Export share in 2015
Sources: Customs Department and Bank of Thailand

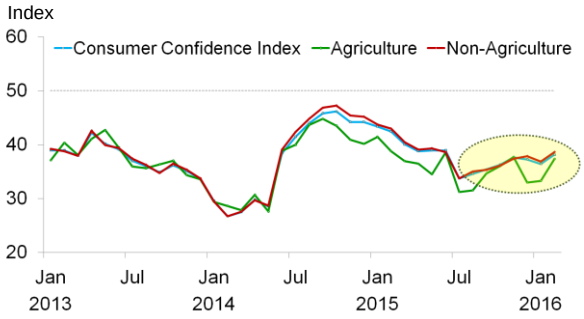
exports of clothing and processed agriculture products to the European Union.

The continuous contraction in exports increasingly affected the real economy as a whole. Moreover, with the effects of the temporary economic boosts started to wane, growth momentum in many sectors, especially consumption and private investment, became weaker.

Private consumption showed signs of moderation after the effects of temporary economic boosts from stimulus measures started to fade. In addition, consumers remained cautious in spending because of concerns about elevated household debt, together with subdued farm incomes that were caused by low commodity prices and China's economic slowdown. Going forward, low input costs from oil prices and improving consumer confidence **(Chart 2.8)** would work to support consumption. Furthermore, labor migration out of the agricultural sector, where income tended to be subdued, into higher-pay sectors would partly support growth of private consumption going forward, especially in the necessity goods category (*Box: The Resilience of Private Consumption in Thailand).*

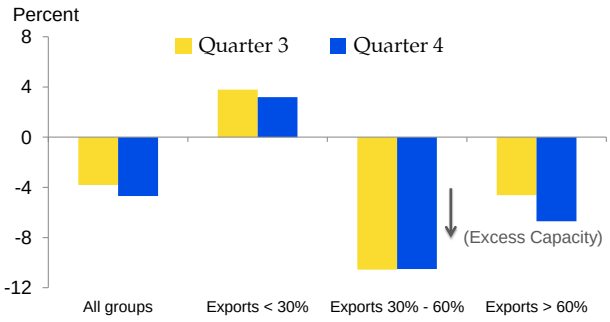
Private investment, especially in the manufacturing sector, remained subdued. A combination of shrinking exports, weak domestic demand, and excess production capacity **(Chart 2.9)** obviated the need for investment to expand production capacity. In addition, foreign direct investment into Thailand's manufacturing sector tended to fall relative to the past **(Chart 2.10)**, particularly in businesses related to electrical appliances, rubber, and plastic. **Nevertheless, investment in services industries, especially those benefiting from government policies, improved to some extent.** Examples included (1)

Chart 2.8 Consumer Confidence Index



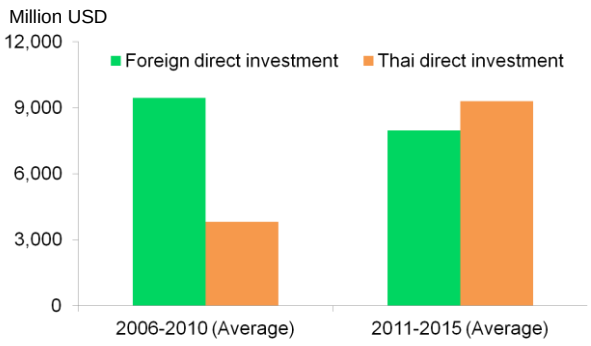
Note: Shares of survey responses are 76.2% being non-agricultural, 11.1% being agricultural, and 12.7% being unemployed/studying.
Source: Ministry of Commerce

Chart 2.9 Capacity utilization in 2015



Note: Difference between current capacity utilization and the average during 2011-2013, excluding period with severe flooding (Oct 2011 – Jan 2012)
Source: Calculations by Bank of Thailand

Chart 2.10 Foreign direct investment and Thai direct investment



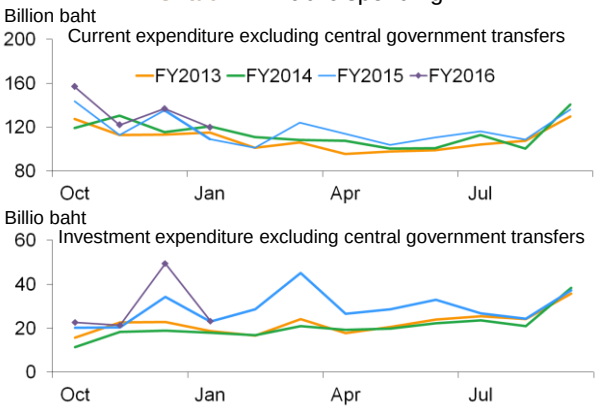
Source: Bank of Thailand

capital investment in telecommunications to support the operation of the 4G mobile network services at the beginning of 2016 and (2) alternative energy investment that was in line with government policy and was intended to support the increased energy need due to growing urbanization.

Public spending and tourism remained key to sustain economic momentum.

Public spending, both current and investment expenditures, continued to be an important driver of growth (Chart 2.11). With the improvement in budget spending efficiency, disbursement of outlays for regular public investment and for projects under the government's phase-2 and phase-3 economic stimulus packages were quickly executed. Meanwhile, disbursement of loans for water resource management and road transportation projects progressed as scheduled. In addition, the government also issued additional stimulus measures during year-end, including the tax deduction measure, whereby taxpayers were allowed to claim an income tax deduction by the amount spent on goods and services up to 15,000 baht, that was aimed at supporting consumer spending and bolstering business confidence.

Chart 2.11 Public spending

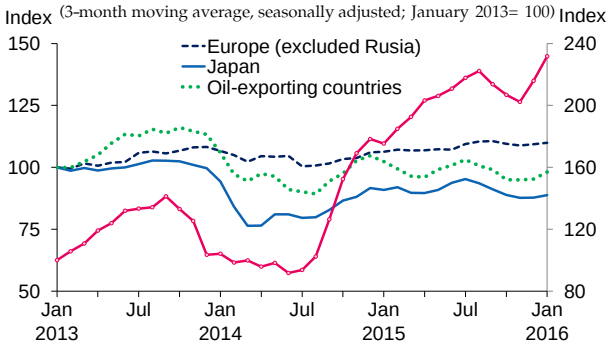


Source: Bureau of Budget; Fiscal Policy Office

Tourism began to show signs of rebound, as reflected in the number of tourists from various countries, and turned out to be a major driver of related services industries such as the wholesale-retail and transportation sectors. Tourism income continued to improve primarily on the back of the increase in Chinese tourists. Despite China's slowdown and yuan depreciation, expansion of low-cost airline routes to Thailand, together with Thailand's popularity as a travel destination, contributed to the surge in

Chinese tourists. Moreover, the number of European tourists rebounded, partly because of an expansion of low-cost airline routes to Thailand. In addition, the number of tourists from commodity-exporting countries, such as Australia, Russia, Indonesia, and Malaysia, also ticked up, although these countries were affected by subdued commodity prices **(Chart 2.12)**. Going forward, tourism was expected to expand solidly given continuing support from low transportation expenses in line with oil prices and from the increases in the number of low-cost airline routes to Thailand.

Chart 2.12 Index of foreign tourists classified by nationality



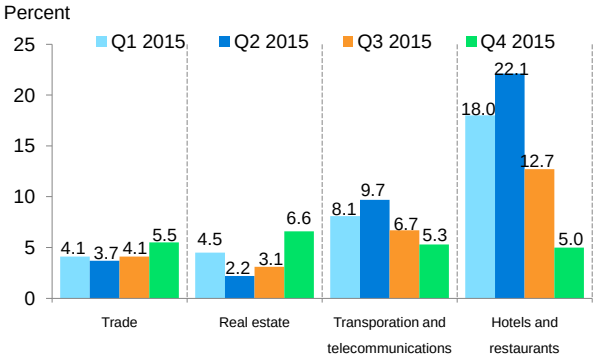
Note: Oil-exporting countries are Australia, Russia, Indonesia, and Malaysia, which account for 20% of total foreign tourists

Source: Department of Tourism

Service industries would play a more significant role in the Thai economy going forward.

Services industries grew strongly in all sectors (Chart 2.13) due mainly to the improvement in the tourism outlook and overall domestic demand that continued to expand. Moreover, services industries associated with the real estate sector benefited from the government's stimulus measures at the end of 2015.[3/] These measures helped increase the number of transactions and transfers of residential units, in particular low-rise properties that typically commanded higher prices than condominiums, and bolstered confidence among property developers by reducing some oversupply in the market.

Chart 2.13 Growth of services industries

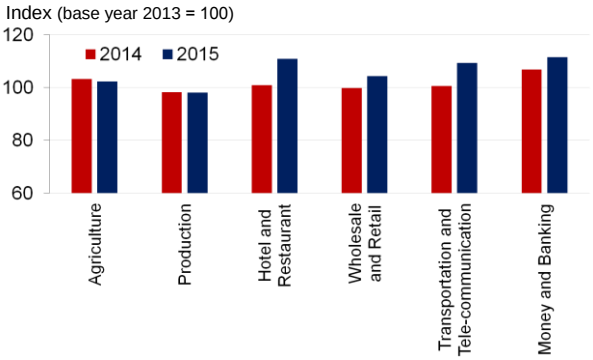


Source : The National Economic and Social Development Board

[3/] The real estate stimulus measures included (1) income tax deduction for first-time home buyers effective from October 13, 2015; (2) loans with a credit line of up to 3 million baht provided by the Government Housing Bank for low- and middle-income households effective from October 19, 2015; and (3) a reduction in transfer and registration fees from 2 percent and 1 percent, respectively, to 0.01 percent effective from October 29, 2015.

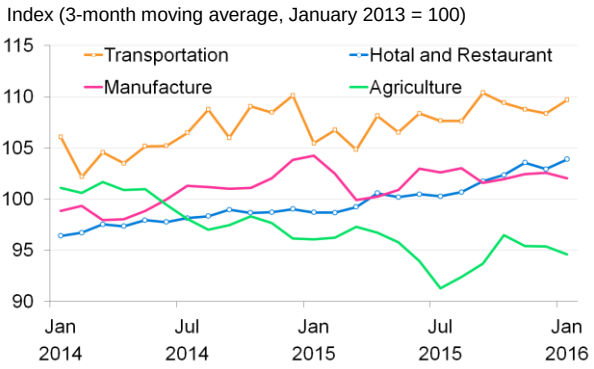
Going forward, services industries would play a crucial role in driving the economy, given that the manufacturing sector was likely to moderate following the slowdown in exports. Moreover, investment in services industries was expected to grow, which would boost total factor productivity in line with rising productivity in modern services sectors, such as transportation, telecommunications, and financial intermediation, which displayed largest gains in productivity per person than other sectors **(Chart 2.14)**. Services industries also played a part in absorbing labor migration out of the agricultural and manufacturing sectors, as tourism-related sectors continued to show strong growth and required only minimal labor skills **(Chart 2.15)**. Services industries would therefore be pivotal in supporting the economy at this current juncture and would be a key driver propelling the Thai economy in the future.

Chart 2.14 Labor productivity



Source: National Economic and Social Development Board; Bank of Thailand

Chart 2.15 Number of employees classified by sector



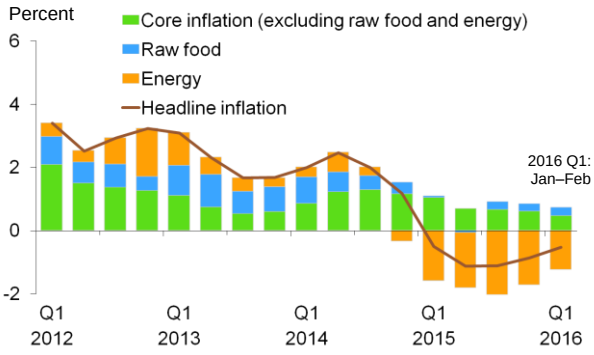
Source: Bank of Thailand

2.3 Production cost and price conditions

Headline inflation remained negative given that global oil prices were still at low levels, while core inflation edged down on account of subdued cost pressures and weaker domestic demand. While the Committee anticipated a rise in headline inflation, which would depend crucially on future oil price developments, it would monitor domestic demand recovery and inflation expectations closely going forward.

Headline inflation remained negative at the beginning of 2016, with the average for the first two months being -0.52 percent. Consumer price inflation was still restrained by subdued domestic energy prices and global oil prices **(Chart 2.16)**. The low crude prices were attributed to (1) an

Chart 2.16 Contribution to headline inflation



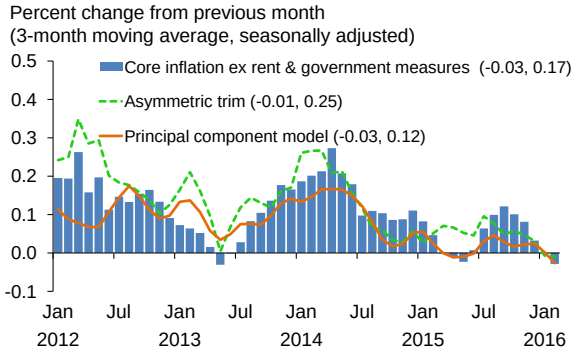
Source: Bureau of Trade and Economic Indices, Ministry of Commerce; calculations by Bank of Thailand

increase in oil production by the Organization of the Petroleum Exporting Countries (OPEC), in particular from Iran after the sanctions were lifted and (2) the drop in oil demand on the back of the slowdown of the global economy, in particular China. Nonetheless, headline inflation was on course to become less negative thanks to the dissipation of the base effect and the increases in fresh food prices, the latter of which were due to drought in some parts of the country that caused prices of vegetables and fruits to edge higher.

Core inflation for the first two months of 2016 averaged at 0.63 percent, slightly edging lower on the back of a weakening domestic economy and subdued cost pressures. While an increase in the excise taxes for tobacco products and automobiles pushed up inflation somewhat, it failed to fully offset the impacts of sluggish demand. Indicators of underlying inflation all showed steady downward trends **(Chart 2.17)**.[4/]

Given the fall in energy prices in tandem with crude prices, production costs of goods and services in the core consumer price basket also declined, as reflected in the decreases in prices of core consumer price goods for which oil was a

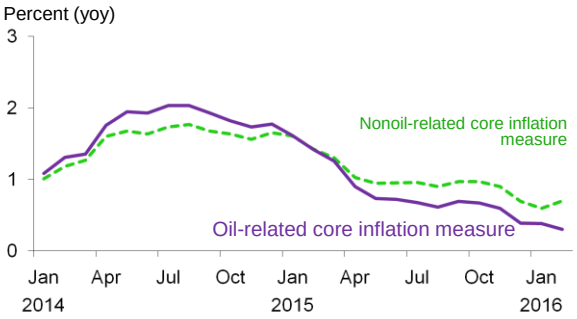
Chart 2.17 Underlying inflation indicators



See footnote 4 for details of each indicator.

Sources: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Chart 2.18 Oil-related core inflation measure



Note: Calculated using weights based on the information in the 2010 Input-Output Table. See footnote 5 for details.

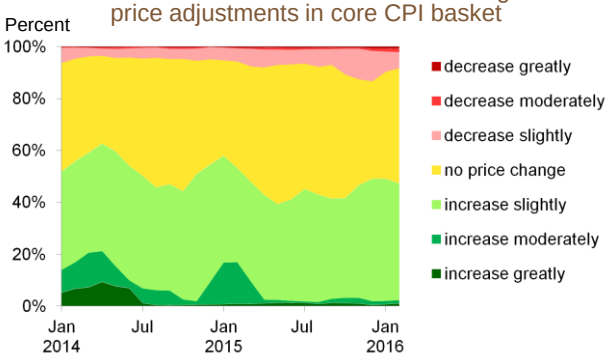
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

[4/] For the legend in Chart 2.17, the number in the brackets on the left is displayed in terms of %MoM (sa, 3mma) as of February 2016, while the number on the right is the long-term average over 2005-2014. *Core inflation ex rent & government measures* excludes (1) rent, which displays high persistence, and (2) changes in prices resulting from government measures and taxes. *Asymmetric trim* excludes goods and services with most volatile price changes, removing from the distribution of price changes the bottom 7 percentile and the top 5 percentile, in order to obtain an inflation trend. *Principal component model* calculates changes in the common statistical factor that drives price movements across categories of goods and services.

production input[5/] **(Chart 2.18)**. Meanwhile, subdued domestic demand also played a role in lowering the prices of goods for which oil was not an input. In addition, businesses found it difficult to raise prices, as reflected in the proportion of goods in the core consumer price basket, whose prices had been raised over the past two years, was on a declining trend **(Chart 2.19)**.

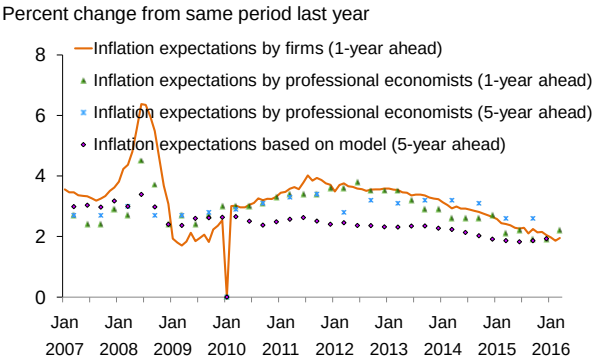
Although inflation expectations edged up slightly in line with recent developments in global crude prices, they remained on a downward trend. One-year-ahead inflation expectations according to surveys of professional economists and businesses respectively stood at 2.2 percent and 1.95 percent as of March 2016. Meanwhile, medium-term inflation expectations remained near the inflation target despite having edged down somewhat **(Chart 2.20)**. Looking ahead, the Committee would continue to monitor the recovery of domestic demand and developments of inflation expectations closely.

Chart 2.19 Shares of items with different degrees of price adjustments in core CPI basket



Note: Calculated from %MoM change according to weight in core CPI basket
Source: Ministry of Commerce, calculations by Bank of Thailand

Chart 2.20 Inflation expectations



Sources: Business Sentiment Survey of Bank of Thailand, Consensus Forecast, and calculations based on macro-finance term structure model

[5/] Two measures of core inflation, one measure with oil used as a direct or indirect input for each item in the core CPI basket and the other measure consisting of goods unrelated to oil, are calculated by assigning weights based on the information in the 2010 Input-Output Table.

Table 2.1 Quarterly inflation

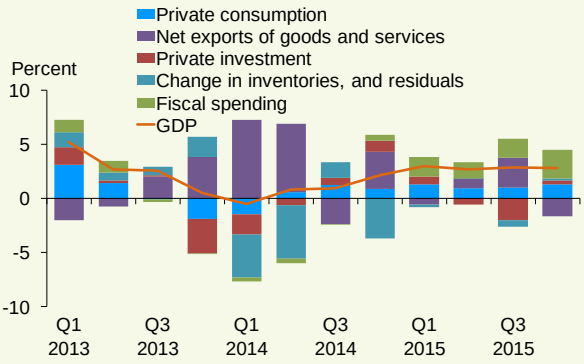
Unit : Percent	2014	2015				2016
		Q1	Q2	Q3	Q4	Jan-Feb
Percentage change from previous year (%△yoy)						
- Headline Consumer Price Index	1.89	-0.50	-1.12	-1.11	-0.86	-0.52
• Core Consumer Price Index	1.59	1.47	0.96	0.92	0.85	0.63
• Raw food	3.46	0.30	-0.40	1.48	1.45	1.62
• Energy	1.68	-12.95	-14.16	-16.42	-14.63	-11.50
Percent change from previous quarter (%△qoq_sa)						
- Headline Consumer Price Index	-	-0.8	-0.1	0.0	0.1	-
• Core Consumer Price Index	-	0.3	0.1	0.3	0.2	-
• Raw food	-	0.0	-0.3	1.3	0.5	-
• Energy	-	-9.0	-0.8	-4.0	-2.1	-

Source: Bureau of Trade and Economic Indices, Ministry of Commerce.
Calculations of percentage change from last quarter, seasonally adjusted, by Bank of Thailand

The resilience of private consumption in Thailand

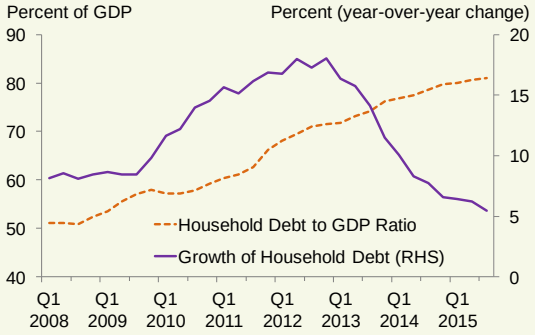
The resilience of private consumption is a critical issue in the assessment of the economic outlook. This is particularly relevant at the current juncture, where exports are not able to act as the main growth driver as in the past due to both cyclical and structural headwinds, and private investment remained subdued. Given such context, Thailand's growth momentum therefore depends critically on public and private consumption, as reflected in their contribution to economic growth **(Chart 1)**.

Chart 1 Contribution to Thailand's GDP growth
(Change from the same period last year)



Source: Office of the National Economic and Social Development Board, calculated by the Bank of Thailand.

Chart 2 Household debt to GDP ratio and growth of household debt



Source: Bank of Thailand

In assessing the strength of private consumption going forward, both negative and positive factors driving consumption are considered as follows. Private consumption was weighed down by several factors last year. First, merchandise exports contracted by 3.4 percent in 2015 from a year earlier due to a slowdown in trading partners' economies and structural limitations. This had a direct repercussion on production and employment in the export-oriented manufacturing sectors. Second, drought and persistently low farm prices weighed on farm incomes, which recorded a contraction of 11 percent in 2015 from a year earlier, and therefore weakened purchasing power of agricultural households. Third, the high level of household debt as a share of GDP continued to be a drag on consumption. The buildup of household debt accelerated during the government's first-car tax rebate scheme in 2012, albeit decelerating since 2013 **(Chart 2)**.

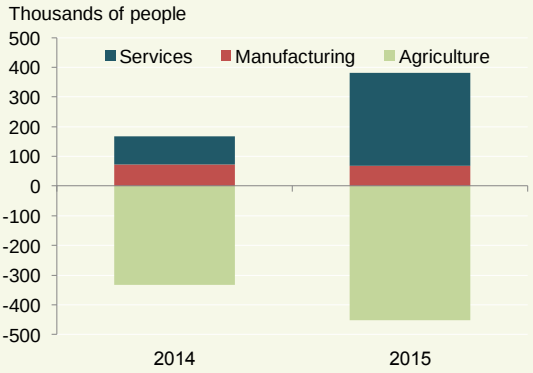
Despite the above negative factors, private consumption still grew thanks to the following factors.

1) Thailand's labor market is flexible. Although drought and depressed farm prices led to a fall in employment in the agricultural sector by 450,000 positions last year, a portion of those labor were able to move into the manufacturing and services sectors, as reflected in an increase of 380,000 positions in the nonagricultural sector. The services sector accounted for the majority of the increase **(Chart 3)**. The remainder is due to the domestic-oriented manufacturing sector, while the export-oriented manufacturing sector contributed only marginally[1]. Within the services sector, the tourist-related services continued to expand solidly thanks to a growing number of tourists **(Chart 4)**. Apparently, Thailand's labor market is flexible enough to absorb labor movements out of the affected sectors, namely the agricultural

[1/] Refers to industries which export at least 30 percent of their total production.

sector and the export-oriented sector. Unemployment thus remained stable during those periods.

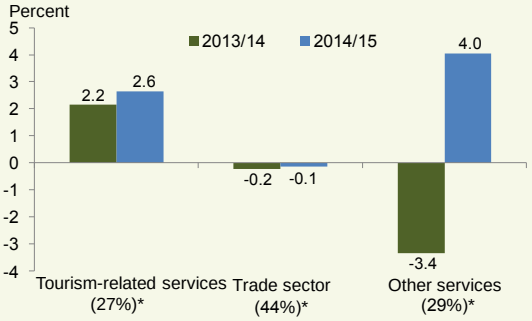
Chart 3 Changes in sectoral employment



Note: Total employment in 2015 is 38 million people

Source: National Statistics Office, and calculated by the Bank of Thailand

Chart 4 Employment growth in Thailand's services sector
(Change from the same period last year)



Note: *denotes a share in total employment in the services sector in 2015 (14 million people)

Source: National Statistics Office, and calculated by the Bank of Thailand

Chart 5 Growth of real income in non-agricultural sectors
(Change from the same period last year)



Source: National Statistics Office, and calculated by the Bank of Thailand

2) Average real income has increased. After the new daily minimum wage of 300 baht became effective on January 1, 2013, nonfarm income in real terms rose by an average of 1 .7 percent per year. Although this growth rate is considered low by historical standards, it has helped shore up overall purchasing power amid subdued farm incomes **(Chart 5)**.

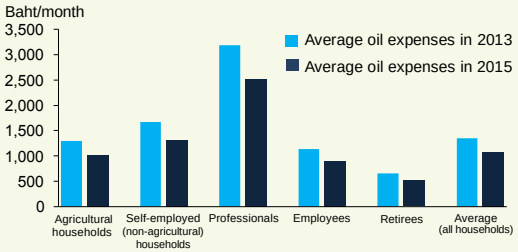
3) A decline in oil prices has partly sustained household purchasing power. With oil-related expenses accounting for 6 percent of household income **(Chart 6)**, a fall in oil prices[2] during 2013-2015 by 27 percent led to a reduction in oil-related expenses by one fifth **(Chart 7)**.

Chart 6 Shares of oil-related expenses in household income in 2013

(Percent of total disposable income)	
Agricultural households	6.6
Self-employed (non-agricultural) households	5.8
Professionals	5.9
Employees	5.4
Retirees	4.3
All households	5.7

Source: Household Socio-Economic Survey

Chart 7 Energy-related expenses by occupation*



Note: *1) Excluding LPG, NGV, and public transportation costs

2) Average oil expenses in 2015 is calculated by the average petrol prices in 2015, multiplied by the amount used by a representative household in each group, assuming that the consumption pattern remains unchanged across time.

Source: Household Socio-Economic Survey

[2/] Proxied by gasohol 91 price.

4) Government measures and improved consumer confidence supported last year's consumption. During the last week of 2015, the government introduced a tax deduction scheme applicable to purchases of goods and services up to 15,000 baht per person. This measure helped shore up confidence of both households and businesses and buoyed private spending during year-end. Such stimulative effect on consumption, however, was expected to be temporary, as some purchases were brought forward from early 2016. As a result, growth momentum of private consumption in January 2016 declined somewhat, as reflected in a lower reading of private consumption indicators, particularly in the semi-durables category such as clothing and apparels. In the absence of the government's measure, however, spending on semi-durables goods would have still registered an expansion. Moreover, private consumption growth during the last quarter of 2015 was partly attributable to the accelerated automobile purchases prior to a hike in vehicle excise tax effective January 1, 2016.

Looking forward, private consumption should continue to receive support from the following factors. First, employment in the domestic-oriented manufacturing sector should continue to expand. Second, employment in the services sector, particularly in the tourist-related services industries, is expected to grow on the back of a positive tourism outlook in 2016. Third, the government's mega investment projects, which are expected to launch this year, should help boost employment in related businesses. Fourth, the prospect of farm incomes should improve as drought is expected to ease in the second half of 2016. Nonetheless, there remain two key factors which will act as a drag on consumption. First, the elevated debt level has put households in a vulnerable position. Second, a slower-than-expected global recovery will likely weigh down on merchandise exports, with further repercussions on the incomes of domestic households and businesses. The Committee will continue to closely monitor these risks to private consumption going forward.

Monetary and Financial Stability

3. Monetary and Financial Stability

Volatility in the financial markets increased due to the recent monetary policy conduct among major industrial countries and growing concerns over the global economic slowdown. However, monetary conditions in Thailand were considered to remain accommodative. Money market rates were broadly unchanged in line with the policy interest rate. Costs of financing in corporate bond markets generally declined from the levels observed in the previous quarter consistent with medium- and long-term government bond yields. Meanwhile, the baht appreciated since last quarter largely because of external influences.

Risks to macro-financial stability rose. The sluggish economic recovery adversely affected income and debt service ability of the household and corporate sectors, in particular, farm households and small- and medium-sized enterprises. While loan quality of the banking sector consequently deteriorated, risks were judged to be contained because of commercial banks' high levels of capital buffers and loan loss provisions and because of large enterprises' strong balance sheets. In the equity and bond markets the search-for-yield behavior was found to still persist given low interest rates. With regard to the real estate sector, risks of oversupply diminished somewhat thanks to the year-end boost from the government's stimulus measures aimed at expediting transactions and transfers. Meanwhile, the government's fiscal position remained sound.

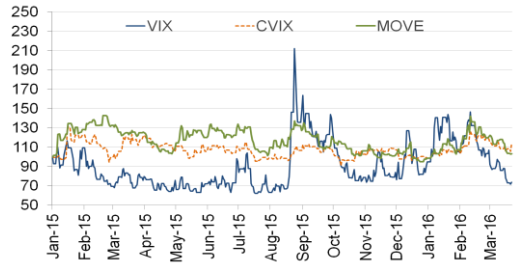
3.1 Financial markets

Global financial markets

Volatility rose in the period since the previous quarter as shown by increases in a few volatility measures (chart 3.1), particularly the VIX index.[1/] The heightened volatility was due to the following key factors.

Chart 3.1 Volatility indices

Index (January 1, 2015 = 100)



VIX is an indicator for stock market volatility, measured by implied volatility of option prices of the S&P 500. CVIX is an indicator for currency volatility, measured by implied volatility of 9 major currencies MOVE is an indicator for bond market volatility, measured by implied volatility of U.S. treasury bonds.

Source: Bloomberg

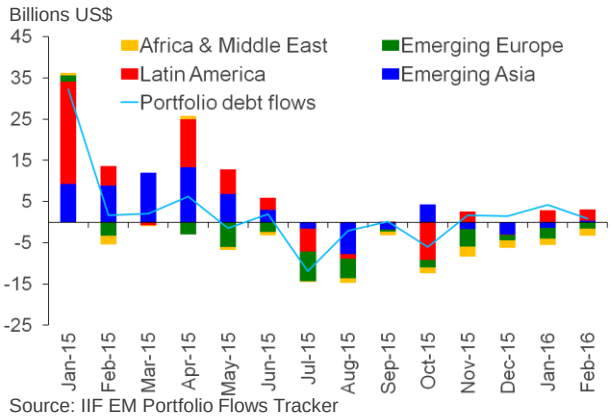
[1/] VIX (Volatility Index) is an indicator for stock market volatility, measured by implied volatility of option prices of the S&P 500.

(1) Concerns about the delay in the global economic recovery increased, with more visible impacts of a slowdown in emerging market economies on advanced economies.

(2) Increasingly accommodative monetary policy was adopted by advanced economies in response to a slower-than-expected recovery prospect, with the Federal Reserve communicating more clearly about a gradual pace of interest rate increases, the Bank of Japan introducing a negative interest rate policy for the first time, and the European Central Bank easing monetary policy more than expected on March 10, 2016.

These factors consequently caused greater fluctuations in international capital flows and led to a return of capital flows to emerging market economies, especially into the bond markets **(chart 3.2).**

Chart 3.2 Total portfolio flows (equity and debt)

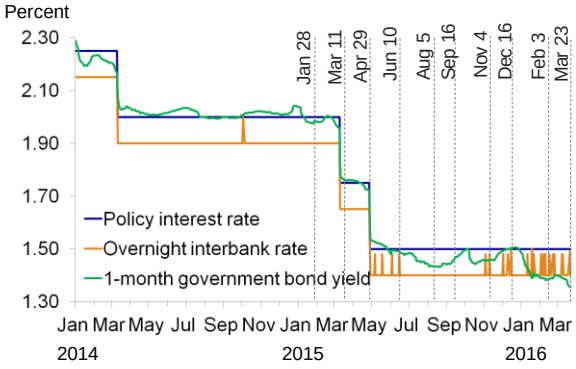


Source: IIF EM Portfolio Flows Tracker

Domestic financial markets

Although short-term interest rates moved closely around the policy rate in the period since the previous quarter, medium- and long-term government bond yields declined significantly. Investors were concerned about the pace of the global recovery that was slower than expected, the softened growth momentum of the domestic economy that was more evident, and the increasingly accommodative monetary policy adopted by advanced economies.

Chart 3.3 Money market interest rates



Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA)

Money, bond, and stock markets

While short-term money market rates moved closely around the policy interest rate between January and mid-March 2016 **(chart 3.3)**, medium- and long-term government bond yields steadily fell **(chart 3.4)**. The decline in yields was driven by investors' concerns about the slower-

Chart 3.4 Government bond yields



Source: Bank of Thailand and Thai Bond Market Association (Thai BMA)

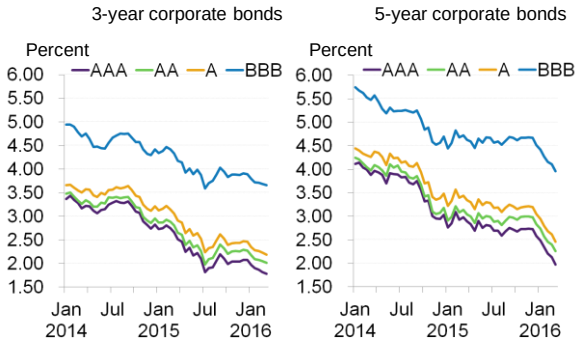
than-expected growth outlook of the global economy and the more apparent signs of the domestic economy's weaker momentum. In addition, the prospect of further monetary accommodation by advanced economies partly contributed to a return of capital flows into the bond markets of emerging markets, including Thailand. Moreover, insufficient domestic bond supply with respect to demand also caused a significant decline in bond yields. Consequently the private sector's costs of borrowing in the bond markets fell in tandem with government bond yields, given that credit spreads between corporate and government bond yields remained largely unchanged **(Chart 3.5**).

The Stock Exchange of Thailand (SET) index edged higher somewhat since the previous quarter, after the net foreign selling pressures subsided and nonresidents' net purchases finally turned positive in February 2016 **(Chart 3.6)**. **Meanwhile, the Market for Alternative Investment (MAI) index began to stabilize**, having steadily declined since the Securities and Exchange Commission (SEC) implemented measures to curb irregular trading activities at the start of 2015. In the first quarter of 2016 retail investors' net purchases importantly supported the MAI index as opposed to their net sales in the last quarter of 2015 **(Chart 3.7)**.

Foreign exchange markets

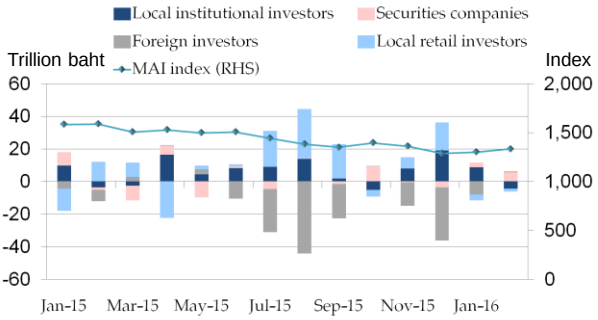
The return of capital flows into the bond and equity markets of reginal economies, including Thailand, caused the baht to appreciate against the U.S. dollar relative to the end of last quarter. Although the strengthening of the baht was in line with that of regional currencies, the extent of the baht appreciation was relatively more on account of local gold traders' increased U.S.

Chart 3.5 Thai corporate bond yields



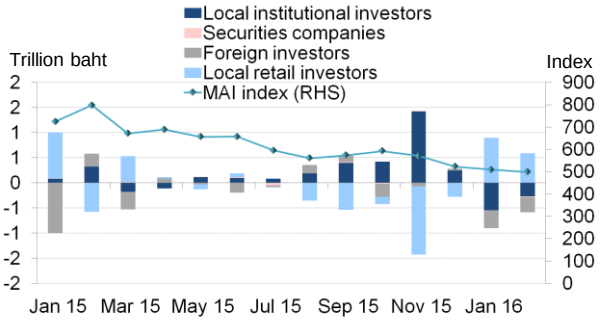
Source: Thai Bond Market Association (Thai BMA)

Chart 3.6 Stock Exchange of Thailand Index and net buy classified by investor types



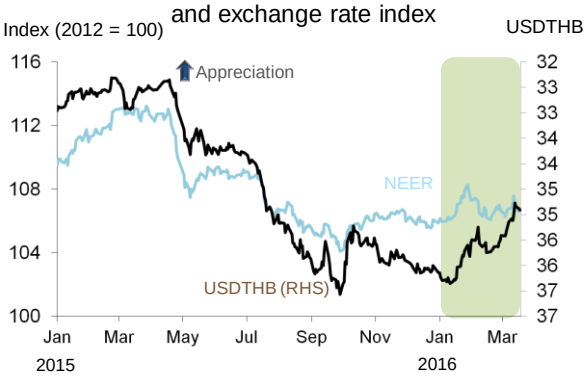
Source: Stock Exchange of Thailand

Chart 3.7 Market for Alternative Investment (MAI) and net buy classified by investor types



Source: Stock Exchange of Thailand

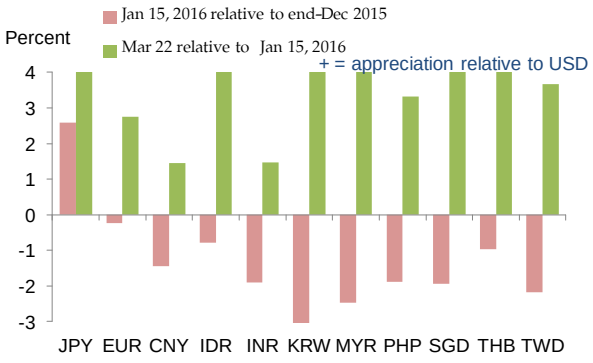
Chart 3.8 Movements in USDTHB and exchange rate index



Source: Bank of Thailand (latest data on March 22, 2016)

dollar selling as gold prices climbed. Nevertheless, the baht occasionally weakened partly because of better-than-expected U.S. economic indicators in both trade and manufacturing sectors that caused the dollar to strengthen. As of market close on March 15, the exchange rate stood at 35.12 baht against the dollar **(Chart 3.8 and 3.9)**.

Chart 3.9 Currency movements relative to USD



Source: Bank of Thailand (latest data on March 22, 2016)

3.2 Financial institutions

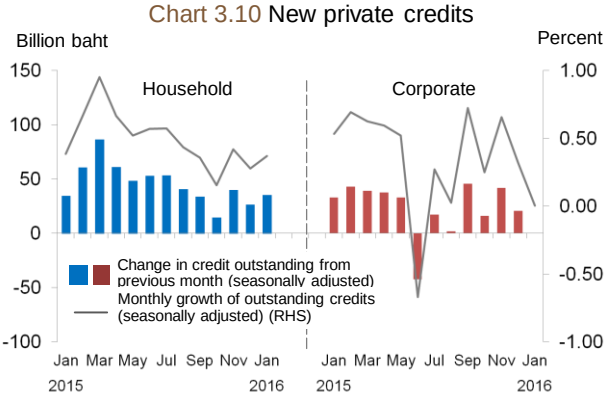
Although the private sector's costs of funds and returns on bank deposits remained low in line with the policy rate, businesses saw little need to raise funds given the sluggish economic outlook. Nonetheless, new loans increased slightly from the third quarter of 2015 largely because of the low interest rate loan program initiated by the government to support SMEs. Meanwhile, the gradual economic recovery also diminished the need for deposit mobilization, as reflected in the decline in issuance of special deposit products.

Interest rates, credits, and deposits

Commercial banks' lending and deposit interest rates remained mostly unchanged throughout the last quarter of 2015 and continuing into March 2016, consistent with the policy rate having been kept unchanged at 1.50 percent since the MPC meeting on June 10, 2015 **(Table 3.10)**.

New loans to households between November 2015 and January 2016 were higher than the amount extended by financial institutions in the third quarter of 2015 **(Chart 3.10).** This increase in new loans was partly due to the government's real estate stimulus measures and accelerated car purchases prior to the 2016 excise tax hike, both of which were temporary factors that

Chart 3.10 New private credits



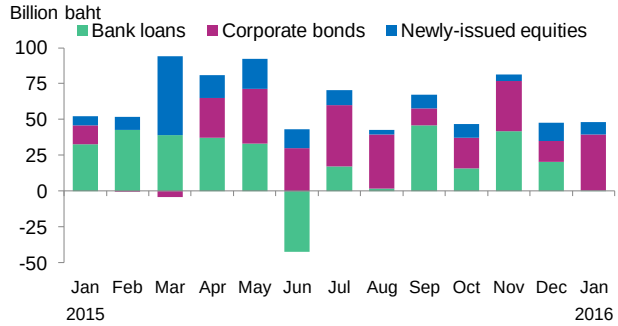
Note: New private credits represent seasonally-adjusted changes in credits outstanding of Other Depository Corporations; Source : Bank of Thailand

caused mortgage and car loans to rise. Nevertheless, according to the credit conditions survey for the fourth quarter of 2015, financial institutions remained rather cautious in extending new credits due to concerns about household repayment ability given the sluggish economic recovery and elevated household debt.

The amount of **total funds raised by the business sector** between November 2015 and January 2016 edged up relative to the amount raised in the third quarter of 2015 **(Chart 3.11)**. This was due primarily to the 100 billion baht low-interest loan program for SMEs. New credits extended by commercial banks were concentrated in the food production, wholesale-retail, and construction sectors, while the outstanding credits for the financial sector and the property sector declined. **However, after the end of the low-interest loan program in January 2016, business funding from commercial bank loans showed some signs of weakening**. SMEs remained subject to tight lending standards, as indicated in the credit conditions survey in the fourth quarter of 2015. Incidentally, the amount of funds businesses raised in the bond markets between November 2015 and January 2016 dipped somewhat from the third quarter of 2015, with large corporations, especially those in the telecommunications, food production and alternative energy industries, continuing to opt for bond issuance in place of bank loans.

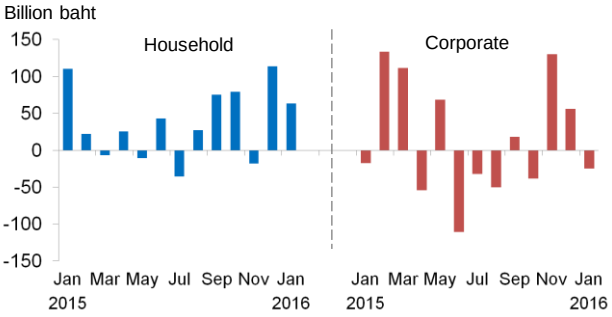
Deposits outstanding at the end of January 2016 increased from the end of October **2015** attributable to both the household and business sectors. Nonetheless, growth in deposits (including bills of exchange, B/E) remained sluggish **(Chart 3.12)**. Given credit growth had remained, and was expected to be, flat,

Chart 3.11 Corporate financing*



Note: *Monthly change in corporate loans (seasonally adjusted), corporate bonds excluding commercial banks and newly issued equities.
Source: Bank of Thailand

Chart 3.12 New deposits*

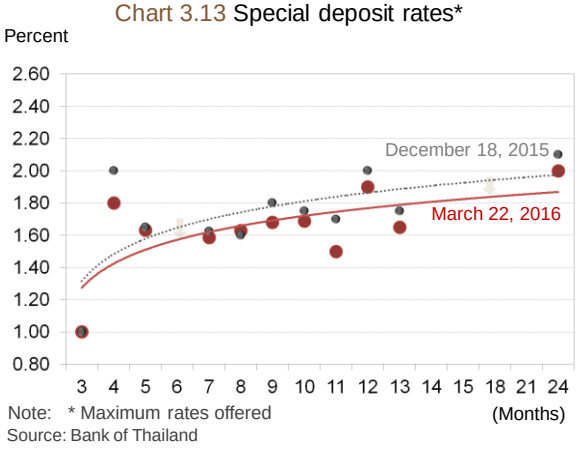


Note: * Change in outstanding deposits at depository institutions (excluding Bank of Thailand). The amounts only account for new deposits, not including transfers within and between commercial banks or rolled-over deposits.

commercial banks saw little need to raise new deposits, as reflected in a continued decline in interest rates on special deposit products **(Chart 3.13)**.

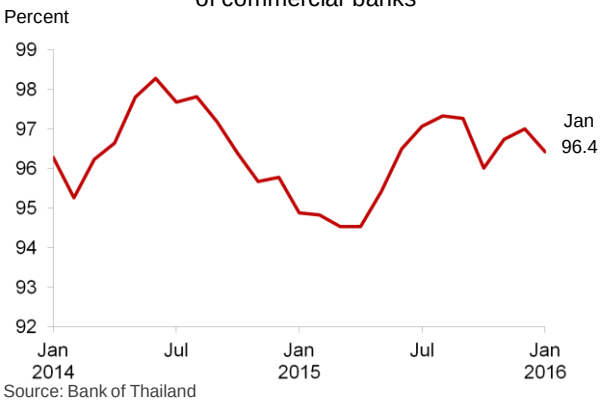
Overall, the **loan to deposit (including B/E) ratio** did not change significantly from the previous quarter, but it recently edged down somewhat due primarily to slowing credit growth **(Chart 3.14)**.

Chart 3.13 Special deposit rates*



Note: * Maximum rates offered
Source: Bank of Thailand

Chart 3.14 Ratio of loans to deposits and B/Es of commercial banks



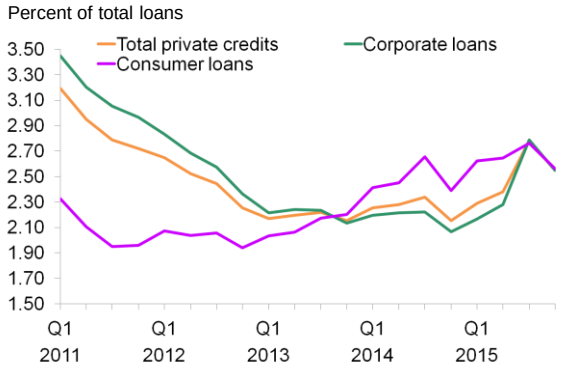
Source: Bank of Thailand

Macro-financial stability

The weak economic recovery increased risks to macro-financial stability as reflected in deteriorating loan quality. Nevertheless, the stability of financial institutions remained assured: commercial banks closely monitored credit quality and continued to hold sufficiently high levels of loan loss provisions and capital buffers that could provide cushion against a decline in loan quality. Meanwhile, prolonged low interest rates set the stage for the search for yield behavior, but risks to financial stability were assessed to be limited.

Loan quality of commercial banks in 2015 deteriorated from 2014 on the back of the sluggish economy. Despite a slight improvement in the fourth quarter of 2015 compared with the third quarter due to commercial banks' effective management of nonperforming loans (NPLs) toward the end of the year, as reflected in the ratio of NPLs to total loans that fell to 2.6 percent at the end of the fourth quarter of 2015 compared with 2.8 percent at the end of the third quarter, the end-2015 NPL ratio was still higher than 2.2 percent at end-2014 **(Chart 3.15)**. This upward NPL trend was due to the deterioration in the loan quality of large corporations and SMEs, in particular those in the retail-wholesale sector affected by the economic slowdown and those in the manufacturing sector further restrained by

Chart 3.15 Non-performing loans of commercial banks



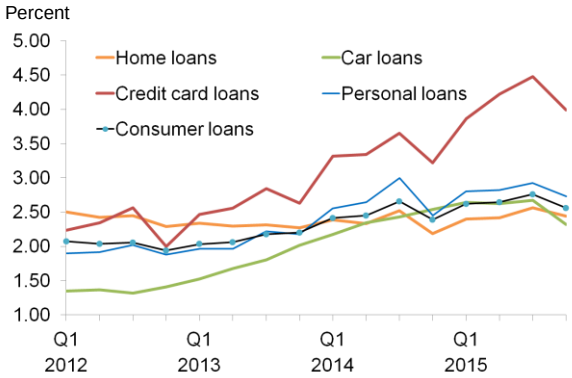
Source: Bank of Thailand

depressed commodity prices. In addition, the NPL ratio for consumer credits in the final quarter of 2015, despite declining in all categories from the previous quarter, was still higher than the end-2014 level **(Chart 3.16)**.

Nevertheless, financial institutions remained sound as commercial banks continued to closely monitor loan quality. For instance, banks increased their efficiency in monitoring and collecting debt to prevent a further rise in NPL ratios. At the same time banks continued to maintain sufficiently high levels of capital buffers and loan loss provisions, as reflected in the actual/regulatory loan loss provision ratio at 156.3 percent, together with the capital adequacy ratio at 17.4 percent that was well above the regulatory required level of 8.5 percent. Looking ahead, the Committee would closely watch loan quality, particularly of SME loans and consumer loans, which could deteriorate owing to the sluggish economic recovery and could therefore affect commercial banks' performance and profitability.

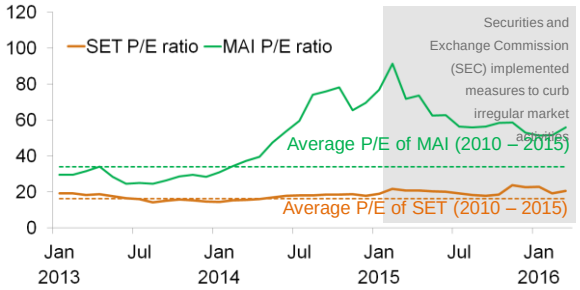
The prolonged low interest rate environment partly encouraged the search for yield behavior observed in the domestic equity and bond markets. Regarding the stock markets, such yield-seeking behavior was reflected in the current and forward price-earnings (P/E) ratios in the MAI, which continued to trade well above historical averages[2] **(Chart 3.17)**. Nevertheless, risks to the stability of the financial system remained limited as equity investors mostly conducted their transactions in cash. Regarding the bond markets, persistently low yields continued to attract the corporate sector to debt

Chart 3.16 Share of non-performing loans in consumer loans



Source: Bank of Thailand

Chart 3.17 Price–Earnings Ratio (P/E)* of SET and MAI



Sources: Stock Exchange of Thailand and Bloomberg (latest data on March 22, 2016)

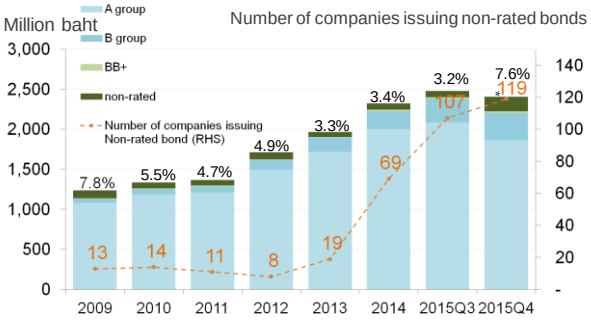
[2] The current and forward P/E ratios of the MAI stood at 56.1 and 24.0, respectively, as of March 22, 2016, well above the 2010-2015 averages of 34.1 and 18.8.

financing. In particular, non-rated bond issuance continued to grow in the fourth quarter of 2015. However, the issuance was of short maturity and only by firms in certain industries **(Chart 3.18)**; in addition, the ratio of non-rated bonds to total bonds outstanding still remained low and the sales of non-rated bonds were restricted to only accredited investors. Therefore, financial system stability risks remained rather limited.

Furthermore, the prolonged low interest rate environment partly led some investors to tolerate a higher level of risks. Notable instances were increases in the relative sizes of the net asset values of daily fixed income funds, property funds / real estate investment trusts, and infrastructure funds. Meanwhile, the size of foreign investment funds remained relatively large **(Chart 3.19)**. Consequently, the above investment funds should be kept under a close watch. Given a sluggish economic recovery and an increase in volatility in the global financial markets, some of these funds could entail higher risks, notably the foreign investment funds, which focused on investing in emerging market economies with low ratings, and fixed income funds,[3] in particular daily fixed income funds, which invested increasingly more in corporate debts.

In addition, given prolonged low interest rates, the Committee would closely monitor the search for yield behavior, including signs of underpricing of risks by investors and financial institutions, which could lead to a buildup of vulnerability in the Thai financial system.

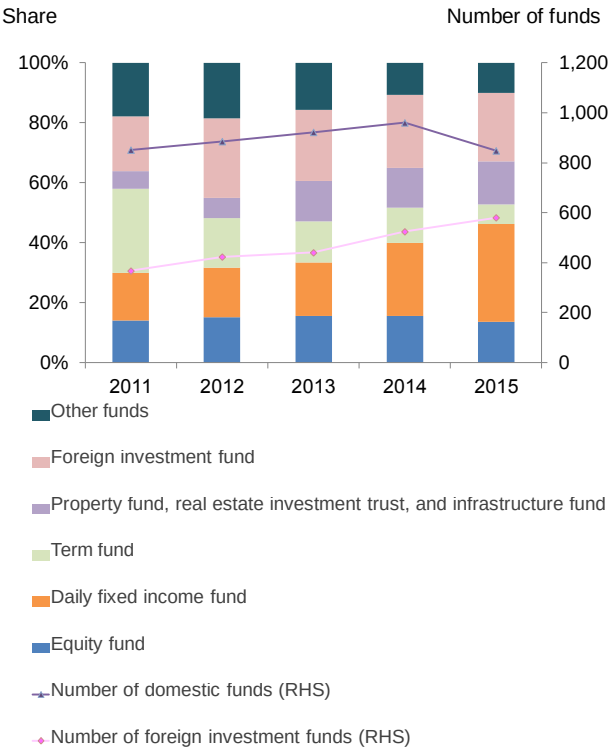
Chart 3.18 Corporate bonds outstanding (short and long maturities)



Note: *Percent of non-rated bonds in total corporate bonds

Source: Thai Bond Market Association (Thai BMA)

Chart 3.19 Mutual funds



Note: Exclude VAYU 1 fund / Country fund / Property fund type 2 – type 4

Source: The Securities and Exchange Commission (SEC)

[3] Fixed income funds consist of three categories, namely money market funds, daily fixed income fund, and term funds.

3.3 Non-financial sectors

Household sector

A slow economic recovery and persistently low commodity prices continued to adversely affect household incomes and debt repayment ability.

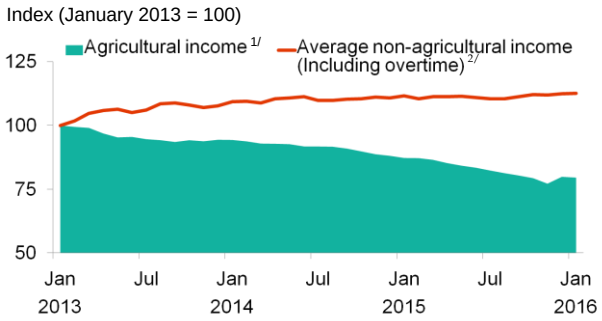
A slow recovery adversely affected household incomes. Farm incomes continued to contract as a result of droughts and low global commodity prices, although non-farm incomes (including overtime pay) gradually increased **(Chart 3.20)**. This led financial institutions to be cautious in approving new loans **(Chart 3.21)** and partly caused a downward trend in household credit growth. As of the third quarter of 2015, household credits expanded by 5.5 percent, lower than the 6.2 percent growth rate recorded in the previous quarter **(Chart 3.22).** Weak household income, coupled with elevated household debt resulting from a rapid rise in leverage in earlier periods, weakened households' debt repayment ability. This was reflected in the increase in the NPL ratio for consumer loans, which rose to 2.6 percent at the end of 2015, up from 2.1 percent at the end of 2014.

Going forward, the outlook for the economic recovery and the results of the government's stimulus measures, particularly the provision of soft loans to farmers affected by droughts and depressed agricultural prices, would need to be closely monitored, as they would have important implications for households' liquidity, income, and debt repayment ability.

Business sector

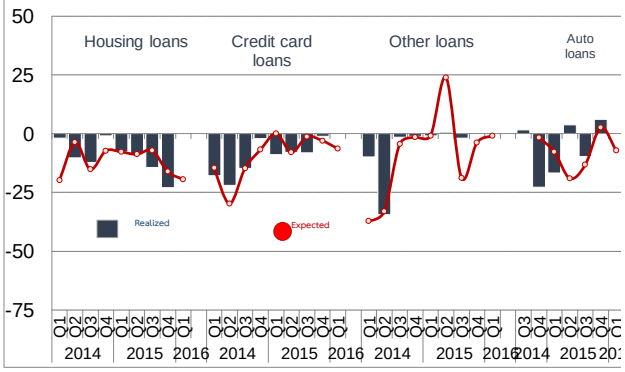
Businesses were negatively affected by the slow economic recovery. Revenues of most

Chart 3.20 Household income



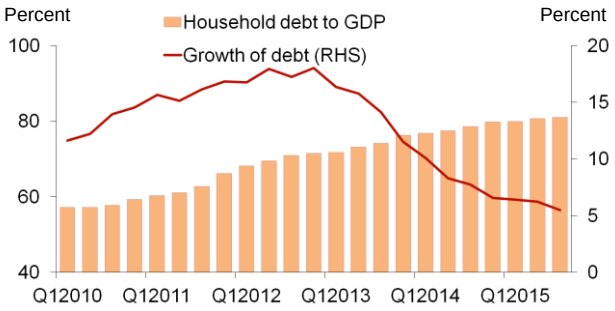
Note: 1/ Seasonally adjusted, 12-month moving average
2/ Seasonally adjusted, 3-month moving average
Sources: Office of Agricultural Economics, National Statistical Office; calculations by Bank of Thailand

Diffusion Index Chart 3.21 Credit standards



Diffusion Index (DI) > 0: easing standards; DI < 0 tightening standards; DI = 0: unchanged

Chart 3.22 Household debt1/ to GDP2/



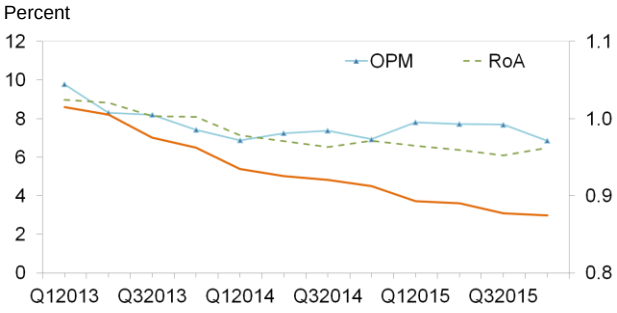
Notes: 1/ Loans to households by financial institutions
2/ Computed using fix-based GDP (base year = 1988)
Source: Bank of Thailand

firms declined from the previous quarter, especially those associated with commodities, for example, the steel and petroleum industry that was hit by falling commodity prices. However, some businesses showed signs of improving, for instance, the property sector that benefited from the government's real estate stimulus measures and the wholesale-retail sector that was supported by the tax deduction measure that allowed taxpayers to claim an income tax deduction by the amount spent on goods and services up to 15,000 baht at the end of 2015.

Falling sales impaired the business ability to generate incomes and profits. Except for the property sector, which turned in greater profits from increasing sales, most businesses experienced falling operating profit margins (OPM) and asset turnover ratios (ATO) **(Chart 3.23)**. Certain industries with declining profitability included the steel industry that was affected by depressed steel prices and the wholesale-retail sector that was particularly hurt by weak domestic demand.

Despite falling sales for most businesses, debt repayment ability improved slightly on balance. This was because most business sectors, especially the petroleum and petrochemical industry, faced lower costs and debt burden, as reflected in an increase in the annualized interest coverage ratio relative to the previous quarter **(Chart 3.24)**. Sectors with low debt burden should thus have sufficient liquidity to cushion against increasing risks. That said, some businesses might have lower debt repayment ability, in particular the wholesale-retail sector. Furthermore, if the listed companies were classified into five groups according to their asset sizes, some firms in the two smallest-sized groups

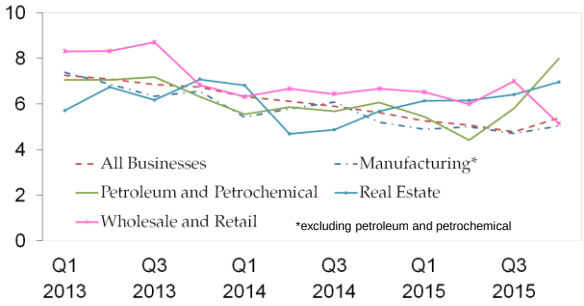
Chart 3.23 Operating profit margin and return on assets



Note: median estimates

Source: Stock Exchange of Thailand; calculation by Bank of Thailand

Chart 3.24 Interest Coverage Ratio (ICR)[1/] classified by firm size[2/]

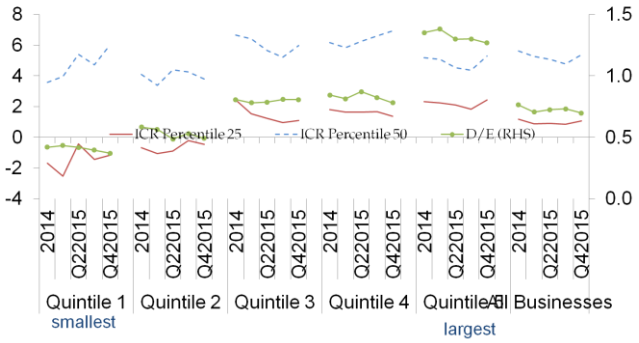


Note: median estimates

Source: Stock Exchange of Thailand; calculation by Bank of Thailand

might experience liquidity problems, as indicated in the apparently much lower levels of liquidity relative to larger businesses **(Chart 3.25).** Furthermore, if these smaller firms were assumed to represent the SMEs that were not traded on the stock exchange, it could be concluded that debt repayment ability by the SMEs had deteriorated, consistent with higher NPL ratios for SMEs as reported at the end of 2015.

Chart 3.25 Interest Coverage Ratio (ICR)* and Debt to Equity (D/E) ratios classified by firm size

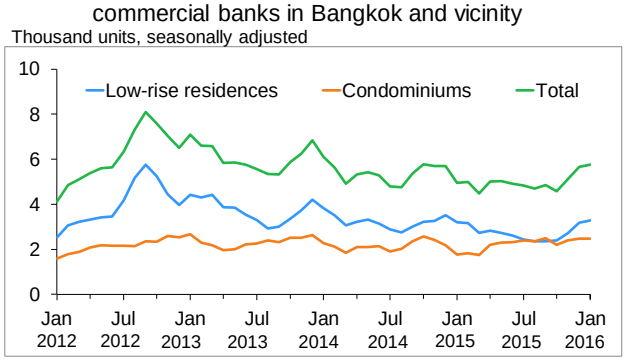


Real estate sector

The property sector in Bangkok and its vicinities improved in the fourth quarter of 2015 on the back of the government's real estate stimulus measures that boosted year-end transactions. However, there remained a high level of excess supply in the property market, which constituted a risk to be monitored.

The property sector in Bangkok and its vicinities improved in the fourth quarter of 2015 on the back of the government's real estate stimulus measures, as reflected in a significant increase in new housing loans, especially for low-rise residences, from commercial banks during November and December 2015 **(Chart 3.26). However, the property sector moderated this year as a whole from a year earlier**, as reflected in both lower demand and supply, consistent with a slow economic recovery, elevated household debt, and tightening lending standards for mortgages. As a result, the proportion of unsold inventory to outstanding supply continued to be high, especially the condominiums priced between 1 and 5 million baht along the purple MRT line **(Table 3.2)**. The large unsold residential units posed a risk to be closely monitored, as the sluggish recovery and a potential delay in the government's infrastructure projects could significantly reduce demand for housing.

Chart 3.26 New residential units with loans from commercial banks in Bangkok and vicinity



Note: *3-month moving average
Source: Agency for Real Estate Affairs (AREA), calculations by Bank of Thailand

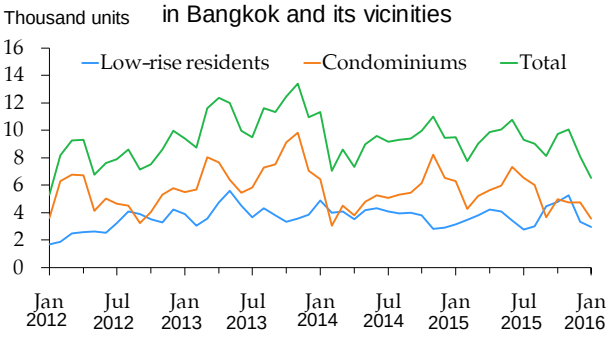
Table 3.2 Shares of unsold inventories at end-2015

Price (million baht)	Central Business District*	Along MRT Green Line**	Along MRT Purple Line***
Under 1	-	-	21%
1 - 2	9%	28%	34%
2 - 3	26%	19%	33%
3 - 5	23%	22%	35%
5 - 10	30%	22%	17%
Above 10	35%	21%	10%

Note: * Phaholyothin Patumwan Sukhumvit Rama4 Yannawa Silom Phranakhon
** Nontaburi Bang-sue Rattanathibet Tha-it Bang-bua-thong
*** Onnuch Srinakarin Bang-na Bearing
Source: Agency for Real Estate Affairs (AREA)

However, the majority of property developers had a strong balance sheet, which would enable them to weather through the impacts from risks of rising unsold inventories. Furthermore, the number of new projects launched during the latter half of 2015 fell sharply for both condominiums and residential houses **(Chart 3.27),** reflecting the adjustment by the property developing in response to the economic outlook, which in part helped reduce the risks of excess supply to some degree.

Chart 3.27 New residential project launches in Bangkok and its vicinities

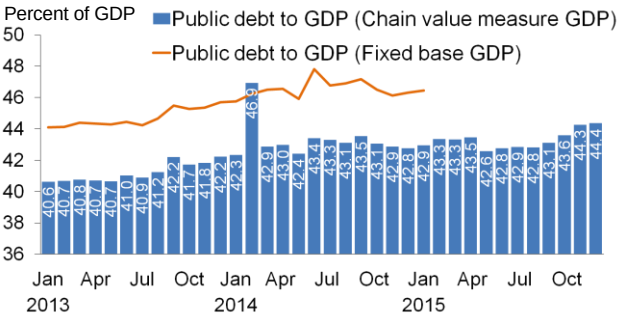


Note: *3-month moving average
Source: Agency for Real Estate Affairs (AREA), calculations by Bank of Thailand

Fiscal sector

Stability of the fiscal sector continued to be strong, as indicated in the public debt to GDP ratio that remained below the fiscal sustainability threshold of 60 percent. As of December 2015, the ratio stood at 44.4 percent, up slightly from 43.1 percent in September 2015 **(Chart 3.28)**. The increase was due to borrowings to cover budget deficit and extra-budgetary investment spending for water resource management and urgent transport system development under the government's second-phase economic stimulus package. Meanwhile, treasury cash balance fell relative to the end of the previous quarter but remained sufficient to meet unexpected fiscal needs. As of the end of January 2016, treasury cash balance stood at 248 billion baht, down from 426 billion baht at the end of September 2015. The decline in the treasury cash balance was due to (1) an increase in regular payments following the pay rise for government officials, (2) a continued increase in outlays for public investment in accordance with the economic stimulus measures, and (3) a fall in government revenues due to the postponement of VAT collection to the following month. The drop in treasury cash balance was not judged to be a

Chart 3.28 Public debt to GDP Ratio*



Notes: (1) Chart shows calendar years, (2) Official figures for debt -to-GDP ratio based on chained -volume measure GDP are available only for February and March 2015. Figures prior to this period are calculated by Bank of Thailand.

concern, as the government front-loaded spending early in the fiscal year, while the revenue collection was scheduled to take place later on.

Going forward, pressures on the fiscal stability would likely increase from (1) additional borrowing needed to cover larger budget deficit, (2) an increase in extra-budgetary borrowings for the transportation infrastructure projects in accordance with the Urgent Transportation Action Plan, according to which a number of projects were expected to be launched in 2016, and (3) additional fiscal liabilities on the back of various quasi-fiscal activities as prescribed by the economic stimulus packages. These additional quasi-fiscal obligations included (1) compensations for the reduction in interest incomes to specialized financial institutions taking part in the soft loan programs and (2) compensations to the Thai Credit Guarantee Corporation for SME loan guarantee fees in accordance with the Portfolio Guarantee Scheme 5 (PGS-5) and also for loan guarantee fees under the second phase of guarantee scheme for micro-entrepreneurs.

Table 3.3 Sectoral Indicators for assessing risks and vulnerabilities to financial stability

Indicators	2014	2015	2014			2015			2016	
			H1	Q3	Q4	H1	Q3	Q4	Jan	Feb
1. Financial markets sector										
Bond market										
Bond spread (10 years-2 years)	1.3	1.1	1.4	1.2	1.4	0.9	1.2	1.1	1.0	0.7
Equity market										
SET Index (End of period)	1,497.7	1,288.0	1,485.8	1,585.7	1,497.7	1,504.6	1,349.0	1,288.0	1,301.0	1,332.4
Actual volatility (SET Index)[1/]	12.5	13.9	14.1	8.4	13.4	12.0	12.3	13.7	21.7	12.5
Price to Earnings Ratio (times)	17.8	22.6	17.9	18.5	17.8	20.1	17.8	22.6	22.9	19.3
FX market										
Actual volatility (baht) (%annualized)[2/]	4.0	5.1	4.3	3.8	3.4	4.5	5.2	6.1	3.6	4.9
Nominal Effective Exchange Rate (NEER)	104.3	108.5	102.7	104.6	107.0	110.8	106.6	105.9	106.5	106.9
Real Effective Exchange Rate (REER)	103.0	104.4	101.9	103.4	105.0	106.99	102.4	101.3	100.7	101.0
2. Financial institutions sector[3/]										
Minimum lending rate (MLR)[4/]	6.75	6.5	6.75	6.75	6.75	6.51	6.5	6.5	6.51	6.51
12-month fixed deposit rate[4/]	1.73	1.4	1.85	1.73	1.73	1.5	1.46	1.4	1.4	1.4
Capital adequacy										
Regulatory capital to risk-weighted asset (%)	16.8	17.4	15.9	17.1	16.8	16.8	17.3	17.4	N.A	N.A
Earnings and profitability										
Net profit (billion baht)	214.1	192.3	111.6	53.9	48.7	103.7	50.0	38.6	N.A	N.A
Return on assets (ROA)	1.2	0.9	1.5	1.3	1.2	1.3	1.2	0.9	N.A	N.A
Liquidity										
Loan to deposit and B/E	95.7	97.0	98.3	97.2	95.7	98.3	97.2	95.7	N.A	N.A
3. Household sector										
Household debt to GDP (times)	79.8	N.A	77.5	78.6	79.8	80.7	81.1	N.A.	N.A	N.A
Financial assets to debt (time)	2.5	N.A	2.5	2.5	2.5	2.5	2.5	N.A.	N.A	N.A
Proportion of loans that are 3 months delinquent (%)										
Thai commercial banks :										
Consumer loans	3.5	2.6	3.5	3.5	3.5	3.5	3.6	2.6	N.A	N.A
Housing loans	1.6	1.7	1.5	1.6	1.6	1.6	1.8	1.7	N.A	N.A
Auto leasing	8.3	7.8	8.2	8.2	8.3	8.5	8.6	7.8	N.A	N.A
Credit cards	2.1	2.2	2.2	2.4	2.1	2.6	2.6	2.2	N.A	N.A
Other personal loans	2.5	2.3	2.4	2.6	2.5	2.5	2.5	2.3	N.A	N.A
4. Non-financial corporate sector[5/]										
Operating profit margin (%)	7.1	7.5	7.1	7.4	6.9	7.8	7.7	6.9	N.A	N.A
Debt to equity ratio (times)	0.8	0.7	0.8	0.8	0.8	0.7	0.7	0.7	N.A	N.A
Income coverage ratio (times)	6.0	5.1	6.2	5.9	5.6	5.2	4.8	5.4	N.A	N.A
Current ratio (times)	1.6	1.7	1.6	1.6	1.6	1.7	1.6	1.7	N.A	N.A

Table 3.3 Sectoral Indicators for assessing risks and vulnerabilities to financial stability

Indicators	2014	2015	2014			2015			2016	
			H1	Q3	Q4	H1	Q3	Q4	Jan	Feb
4. Non-financial corporate sector [5/] (Continues)										
Delinquency and NPL ratio										
Large bussiness	1.1	1.6	1.2	1.1	1.1	2.0	2.0	1.6	N.A	N.A
SMEs	3.1	3.5	3.4	3.4	3.1	3.4	3.6	3.5	N.A	N.A
5. Real estate sector										
demand										
The number of approved mortgages from banks										
(Bangkok and its vicinity)	62,839	59,667	29,195	17,345	16,299	26,503	15,770	17,394	3,104	4,580
Single-detached and semi-detached houses	15,694	13,152	7,537	4,230	3,927	6,384	3,115	3,653	689	1,065
Townhouses and commercial buildings	21,764	19,210	10,705	5,844	5,215	9,199	4,812	5,199	1,071	1,555
Condominiums	25,381	27,305	10,953	7,271	7,157	10,920	7,843	8,542	1,344	1,960
Supply										
The number of new openings (Bangkok and its vicinity)	111,211	107,988	54,576	28,268	28,367	59,316	24,436	24,236	6,669	5,369
Single-detached and semi-detached houses	18,933	17,637	10,411	5,349	3,173	6,354	7,388	3,895	744	1,304
Townhouses and commercial buildings	26,980	27,518	14,760	6,611	5,609	15,349	6,034	6,135	1,312	2,269
Condominiums	65,298	62,833	29,405	16,308	19,585	37,613	11,014	14,206	4,613	1,796
Housing price index [6/]										
Single-detached houses (including land)	127.9	129.3	124.9	128.8	127.9	127.7	129.6	129.3	128.6	128.8
Townhouses (including land)	133.5	137.5	127.4	131.3	133.5	133.7	134.7	137.5	138.3	139.2
Condominiums	151.1	160.9	142.0	141.8	151.1	152.1	156.4	160.9	160.1	159.9
Land	145.6	168.8	140.6	143.6	145.6	151.9	157.8	168.8	169.7	167.1
6. Fiscal sector										
Public debt to GDP (%)	46.3	44.4	47.8	47.2	46.3	42.8	43.1	44.4	N.A	N.A

[1/] Annualized standard deviation of return

[2/] Daily volatility (using exponentially weighted moving average method)

[3/] Based on data of all commercial banks

[4/] Average value of 4 largest Thai commercial banks

[5/] Only listed companies on SET (median); with revisions to data and methodology.

[6/] Due to the fact that the structure of the housing market has changed significantly, the Bank of Thailand is currently improving the price index to better reflect the structure change